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UNITED STATES DEPARTMENT OF COMMERCE

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864-US-CNT

CONFIRMATION NO. 4414

POA ACCEPTANCE LETTER



OC000000102793178

196165
Alleman Hall LLP/MS
900 SW 5th Avenue
Suite 2300
Portland, OR 97204

Date Mailed: 03/13/2026

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 03/03/2026.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/qmjackson/



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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864-US-CNT

CONFIRMATION NO. 4414

141674

Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Suite 1100
Washington, DC 20005

POWER OF ATTORNEY NOTICE



OC000000102793039

Date Mailed: 03/13/2026

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 03/03/2026.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/qmjackson/

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	18/541,216
Filing Date	December 15, 2023
First Named Inventor	Gonzalo A. RAMOS, Jin A. SUH, Shiqian Rachel NG, Christopher Alan MEEK, Haekyu PARK
Title	DOCUMENT RE-FINDING SYSTEM
Art Unit	2161
Examiner Name	OSCAR WEHOVZ
Attorney Docket Number	MST-0561/410864-US-CNT

SIGNATURE of Applicant or Patent Practitioner

Signature	/Mark D. Alleman/	Date (Optional)	
Name	Mark D. Alleman	Registration Number	42257
Title (if Applicant is a juristic entity)	Attorney for Applicant		
Applicant Name (if Applicant is a juristic entity)	Microsoft Technology Licensing, LLC		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

*Total of 1 forms are submitted.

A Federal agency may not conduct or sponsor, and a person is not required to respond to, nor shall a person be subject to a penalty for failure to comply with an information collection subject to the requirements of the Paperwork Reduction Act of 1995, unless the information collection has a currently valid OMB Control Number. The OMB Control Number for this information collection is 0651-0035. Public burden for this form is estimated to average 3 minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the information collection. Send comments regarding this burden estimate or any other aspect of this information collection, including suggestions for reducing this burden to the Chief Administrative Officer, United States Patent and Trademark Office, P.O. Box 1450, Alexandria, VA 22313-1450 or email InformationCollection@uspto.gov. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS.** If filing this completed form by mail, send to: **Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

196165

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number:

OR



Firm or
Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Microsoft Technology Licensing, LLC



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

Name

Tyler Fuller

Title

Vice President

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.



Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	03/03/2026 06:49:41 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	12299058
CONFIRMATION #	4414	FILED BY	EMILY AGAMIE
PATENT CENTER #	74704480	FILING DATE	12/15/2023
CUSTOMER #	141674	FIRST NAMED INVENTOR	Gonzalo A. RAMOS
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	MARK ALLEMAN

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
MS - MTL POA 196165.pdf	2	Power of Attorney	946 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
MS - MTL POA 196165.pdf	7077E594EC756A62A20F0F7B521A83ADC92906E9FB20A8FEF7C5304BA64D3D9902A4205AC352AD4FA8301E67F279253B34861139EF40D69F7C06F99C1E122141

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414
141674	7590	05/13/2025	EXAMINER	
Faegre Drinker Biddle & Reath LLP (Microsoft)			WEHOVZ, OSCAR	
1500 K Street, N.W.			ART UNIT	
Suite 1100			PAPER NUMBER	
Washington, DC 20005			2161	
			NOTIFICATION DATE	DELIVERY MODE
			05/13/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

msdocket@microsoft.com
patentdocketingdvr@faegredrinker.com

APPLICATION NO.	ISSUE DATE	PATENT NO.
18/541,216	13-May-2025	12299058

Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Washington, DC 20005

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	05/13/2025	12299058	MST-0561/410864-US-CNT	4414

141674 7590 04/30/2025
Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Suite 1100
Washington, DC 20005

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Gonzalo A. RAMOS, Kirkland, WA;
Jin A. SUH, Seattle, WA;
Shiqian Rachel NG, Seattle, WA;
Christopher Alan MEEK, Kirkland, WA;
Haekyu PARK, Atlanta, GA;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

Microsoft Technology Licensing, LLC, Redmond, WA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Gonzalo A. RAMOS et al.

Confirmation No.: 4414

Application No. 18/541,216

Art Unit: 2161

Filed: December 15, 2023

Examiner: Oscar Wehovz

For: DOCUMENT RE-FINDING SYSTEM

COMMENTS ON STATEMENT OF REASONS
FOR ALLOWANCE UNDER 37 C.F.R. § 1.104(e)

Mail Stop Issue Fee
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

Dear Commissioner:

Applicant has received the Examiner's Statement of Reasons for Allowance with the Notice of Allowance and Allowability dated January 17, 2025, regarding the above-identified application. Reasons for allowance are only warranted in instances in which "the record of the prosecution as a whole does not make clear the Examiner's reasons for allowing a claim or claims." 37 C.F.R. § 1.104(e). In the present case, Applicant believes the record as a whole makes clear the reasons for allowance and therefore no statement by the Examiner is necessary or warranted. Therefore, the record should reflect that Applicant does not necessarily agree with the statements of reasons for allowance.

For example, the record should reflect that Applicant does not necessarily agree with the Examiner's statements regarding the purported disclosure of several U.S. patent documents and the purported relevancy of the several U.S. patent documents to the claims.

Additionally, Applicant believes the claims are allowable for reasons in addition to those set forth by the Examiner. As an example, although the Examiner refers to specific elements of

the independent claims, Applicant respectfully submits that the claims, including the dependent claims, include other distinguishing recitations.

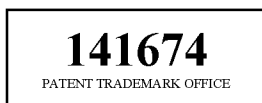
Moreover, each of the claims stands on its own merits and is patentable also because of the combination it recites and not merely because of the presence of a particular subset of elements. Thus, it is respectfully submitted that each claim is allowable also at least because the subject matter defined by its language is neither anticipated by, nor would have been obvious when taken as a whole in view of the art of record.

Should there be any outstanding matters that need to be resolved in the present application, the Examiner is respectfully requested to contact the undersigned. If necessary, the Commissioner is hereby authorized in this, concurrent, and future replies, to charge payment or credit any overpayment to Deposit Account No. 50-0573 for any additional fees required under 37 C.F.R. § 1.16 or under 37 C.F.R. § 1.17.

Respectfully submitted,

Date: April 16, 2025

/Ethan W. Bell/
Ethan W. Bell, Reg. No. 75,549



FAEGRE DRINKER BIDDLE & REATH LLP
1144 15th Street Suite 3400
Denver, Colorado 80202
303 927 5220

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

141674 7590 01/17/2025
Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Suite 1100
Washington, DC 20005

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Irina L. Mikitiouk	(Typed or printed name)
/Irina L. Mikitiouk/	(Signature)
April 16, 2025	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414

TITLE OF INVENTION: DOCUMENT RE-FINDING SYSTEM

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	04/17/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
WEHOVZ, OSCAR	2161	707-722000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Microsoft Technology Licensing, LLC

Redmond, WA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 50-0573

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Ethan W. Bell/

Date April 16, 2025

Typed or printed name Ethan W. Bell

Registration No. 75,549



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION # 18/541,216	RECEIPT DATE / TIME 04/16/2025 03:19:56 PM Z ET	ATTORNEY DOCKET # MST-0561/410864-US-CNT
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Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	70035598	FILING DATE	12/15/2023
CUSTOMER #	141674	FIRST NAMED INVENTOR	Gonzalo A. RAMOS
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Ethan Bell

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-04-16_MST-0561_410864-US-CNT_Issue_Fee.pdf	1	Issue Fee Payment (PTO-85B)	121 KB
2025-04-16_MST-0561_410864-US-CNT_Comments_for_Allowance.pdf	2	Post Allowance Communication - Incoming	97 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-04-16_MST-0561_410864-US-	16FA303F402124AF89D39B7857F552D23C2BDD7AC9AFA79605FC685A7F1A6DD1AAEC47A5F8C2D569C6CA026B30580BC8

CNT_Issue_Fee.pdf

A590A34547333312292098F437F4749

2025-04-16_MST-
0561_410864-US-
CNT_Comments_for_Allowanc
e.pdf

1B94CD140731CDA95E8EBA4FB198A8F69D1B748DB37EEE98
46B2500A67467769D24ED907AE7CAE942864C2D5AC73AFF6
CF525A702E83BBFD4FC8E6D2AE898863

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	04/16/2025 03:19:56 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	70035598	AUTHORIZED BY	Ethan Bell
CUSTOMER #	141674	FILING DATE	12/15/2023
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Gonzalo A. RAMOS

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 5984	E20254FF20498755	Irina Mikitiouk
PRE-AUTHORIZED ACCOUNT	PRE-AUTHORIZED CATEGORY	
500573	37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)	

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1200.00	1	1200.00
TOTAL AMOUNT:				\$1,200.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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Alexandria, Virginia 22313-1450
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414
141674	7590	03/20/2025	EXAMINER	
Faegre Drinker Biddle & Reath LLP (Microsoft)			WEHOVZ, OSCAR	
1500 K Street, N.W.			ART UNIT	
Suite 1100			PAPER NUMBER	
Washington, DC 20005			2161	
			NOTIFICATION DATE	DELIVERY MODE
			03/20/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

msdocket@microsoft.com
patentdocketingdvr@faegredrinker.com

PATENT
Attorney Docket No. MST-0561/410864-US-CNT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Gonzalo A. RAMOS et al.

Confirmation No.: 4414

Application No. 18/541,216

Art Unit: 2161

Filed: December 15, 2023

Examiner: Oscar Wehovz

For: DOCUMENT RE-FINDING SYSTEM

AMENDMENT UNDER 37 C.F.R. §1.312

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

Dear Commissioner:

In connection with the above-identified application, please enter the following amendment:

Amendments to the Claims being on page 2 of this paper.

Remarks begin on page 8 of this paper.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Gonzalo A. RAMOS et al.

Confirmation No.: 4414

Application No. 18/541,216

Art Unit: 2161

Filed: December 15, 2023

Examiner: Oscar Wehovz

For: DOCUMENT RE-FINDING SYSTEM

AMENDMENT UNDER 37 C.F.R. §1.312

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

Dear Commissioner:

In connection with the above-identified application, please enter the following amendment:

Amendments to the Claims being on page 2 of this paper.

Remarks begin on page 8 of this paper.

Amendments to the Claims:

This listing of claims will replace all prior versions and listings of claims in the application.

1. (Previously Presented) A method for assisting users in re-finding documents, the method comprising:

generating, for each concept clip of a set of concept clips, an embedding, wherein the set of concept clips corresponds to a concept searched by a user;

generating a concept embedding based on the embeddings of the set of concept clips;

determining a semantic relationship between the concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and

based on the determined semantic relationship, providing, for display at a client device, an indication of the semantic relationship between the concept and one or more document clips of the one or more document embeddings.

2. (Previously Presented) The method of claim 1, wherein the set of documents comprises a web page in a browsing history of the user.

3. (Previously Presented) The method of claim 1, wherein the document clip includes a text paragraph parsed out from a document of the set of documents or an image from the document.

4. (Previously Presented) The method of claim 1, wherein the document clip includes at least one user-defined clip that includes information not extracted from a document of the one or more documents.

5. (Previously Presented) The method of claim 1, wherein the set of concept clips includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user.

6. (Previously Presented) The method of claim 1, wherein the document embeddings for the document clips were generated using a machine learning model based on content, and wherein the document embeddings capture semantic meaning of the content, and wherein the embeddings for respective concept clips are generated using the machine learning model.

7. (Previously Presented) The method of claim 1, wherein the concept embedding is generated based on a weighted combination of the embeddings generated for the set of concept clips.

8. (Previously Presented) The method of claim 1, wherein the indication of the semantic relationship comprises graphical representations of the one or more document clips in a multidimensional canvas, wherein the graphical representations of the document clips are placed relative to each other based on calculated distances between the one or more document embeddings for the corresponding document clips.

9. (Previously Presented) The method of claim 8, wherein providing the indication of the semantic relationship further comprises:

placing a graphical representation of the concept at a first location in the multidimensional canvas;

determining locations of the graphical representations of the document clips relative to the first location of the graphical representation of the first concept based on calculated distances between the document embeddings and the concept embedding; and

positioning the graphical representations of the document clips in relation to the graphical representation of the concept in the multidimensional canvas in accordance to the determined distances.

10. (Previously Presented) The method of claim 9, wherein the set of concept clips is a first set of concept clips and the concept is a first concept, and the method further comprises:

receiving a second set of concept clips defining a second concept for searching for content of interest to the user;

placing a graphical representation of the second concept at a second location in the multidimensional canvas; and

updating locations of the graphical representations of the document clips such that:

i) graphical representations of document clips that are more semantically similar to the first concept than the second concept are placed closer to the graphical representation of the first concept in the multidimensional canvas, and

ii) graphical representations of document clips that are more semantically similar to the second concept than the first concept are placed closer to the graphical representation of the second concept in the multidimensional canvas.

11. (Currently Amended) The method of claim 10, wherein updating locations of the graphical representations of the document clips comprises:

generating embeddings for respective concept clips in the second set of concept clips;

generating a second concept embedding based on the embeddings generated for the respective concept clips in the second set of concept clips;

determining updated locations of the graphical representations of the document clips in the multidimensional canvas based on:

i) the first location of the graphical representation of the first concept in the multidimensional canvas,

ii) calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding,

iii) the second location of the graphical representation of the second concept in the multidimensional canvas, and

iv) calculated distances between the embeddings generated for the corresponding document clips and the second concept embedding; and

moving the graphical representations of the document clips to the updated locations in the multidimensional canvas.

12. (Previously Presented) The method of claim 10, further comprising:
- detecting that the graphical representation of the first concept has been repositioned to a new location in the multidimensional canvas by the user; and
 - in response to detecting that the graphical representation of the first concept has been repositioned, repositioning graphical representations of the document clips in the multidimensional canvas such that graphical representations of document clips that are more semantically similar to the first concept than the second concept move closer to the new location of the graphical representation of the first concept.
13. (Previously Presented) A system for assisting users in re-finding documents, the system comprising:
- one or more computer readable storage media; and
 - program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to:
 - generate, for each concept clip of a set of concept clips that correspond to a concept searched by a user using a machine learning model, an embedding;
 - generate a concept embedding based on the embeddings of the set of concept clips;
 - determine a semantic relationship between the concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and
 - generate a graphical user interface based on the determined semantic relationship depicting the semantic relationship between the concept and one or more document clips of the one or more document embeddings.
14. (Currently Amended) The system of claim 13, wherein the set of documents comprises [[a]] a web page in a browsing history of the user.

15. (Previously Presented) The system of claim 13, wherein the document clip includes a text paragraph parsed out from a document of the set of documents, an image from the document, or a user-defined clip that includes information not extracted from the document.

16. (Previously Presented) The system of claim 13, wherein the set of concept clips includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user.

17. (Previously Presented) The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the concept embedding based on a weighted combination of the embeddings generated for the set of concept clips.

18. (Previously Presented) The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the graphical user interface at least by:

placing graphical representations of the document clips in a multidimensional canvas in the graphical user interface, wherein the graphical representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips;

placing a graphical representation of the concept at a location in the multidimensional canvas;

determining locations of the graphical representations of the document clips relative to the location of the graphical representation of the concept based on calculated distances between the embeddings generated for the corresponding document clips and the concept embedding; and

positioning the graphical representations of the document clips in relation to the graphical representation of the concept in the multidimensional canvas in accordance to the determined distances.

19. (Previously Presented) A computer storage medium comprising computer executable instructions, which when executed by at least one processor, perform a method comprising:

generating, for each concept clip of a set of concept clips, an embedding, wherein the set of concept clips corresponds to a concept searched by a user;

generating a concept embedding based on the embeddings of the set of concept clips;

determining a semantic relationship between the concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and

based on the determined semantic relationship, providing, for display at a client device, an indication of the semantic relationship between the concept and one or more document clips of the one or more document embeddings.

20. (Previously Presented) The computer storage medium of claim 19, wherein the method further comprises:

placing a representation of the concept at a location in the multidimensional canvas;

determining locations of representations of the document clips relative to the location of the representation of the concept based on calculated distances between the document embeddings and the concept embedding; and

placing the representations of the document clips in relation to the representation of the concept in the multidimensional canvas in accordance to the determined distances.

REMARKS

The above amendments are made following the Notice of Allowance mailed on January 17, 2025. Herein claims 11 and 14 have been amended to correct minor formalistic errors. No substantive issues are raised and no new matter is added. Thus, Applicant requests consideration and entry of the Amendment at the Examiner's earliest opportunity. If the Examiner has any questions, please contact the undersigned at the below-listed telephone number.

If necessary, the Commissioner is hereby authorized in this, concurrent, and future replies, to charge payment or credit any overpayment to Deposit Account No. 50-0573 for any additional fee required under 37 C.F.R. §§ 1.16 or 1.17.

Respectfully submitted,

Date: March 5, 2025

/Ethan W. Bell/
Ethan W. Bell, Reg. No. 75,549



FAEGRE DRINKER BIDDLE & REATH LLP
1144 15th Street, Suite 3400
Denver, Colorado 80202
303 927 5220



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	03/05/2025 04:54:03 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	69454696	FILING DATE	12/15/2023
CUSTOMER #	141674	FIRST NAMED INVENTOR	Gonzalo A. RAMOS
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Ethan Bell

Documents

TOTAL DOCUMENTS: 3

DOCUMENT		PAGES	DESCRIPTION	SIZE (KB)
2025-03-05_MST-0561_410864-US-CNT_Amendment_Under_Rule_312.pdf		8	-	110 KB
2025-03-05_MST-0561_410864-US-CNT_Amendment_Under_Rule_312-A.NA.pdf	(1-1)	1	Amendment after Notice of Allowance (Rule 312)	76 KB
2025-03-05_MST-0561_410864-US-CNT_Amendment_Under_Rule_312-CLM.pdf	(2-7)	6	Claims	85 KB
2025-03-05_MST-0561_410864-US-	(8-8)	1	Applicant Arguments/Remarks Made in an Amendment	94 KB

CNT_Amendment_Under_
Rule_312-REM.pdf

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-03-05_MST- 0561_410864-US- CNT_Amendment_Under_Rule _312.pdf	CBFBF1A7D5A2B658D7326168BCF52803CDFE6DE784560FD9 113AFEBBC84110F773A05EBE49377D0E6778B09E42275838D1 822D9ADE52C6756591CB5AEA211D36B
2025-03-05_MST- 0561_410864-US- CNT_Amendment_Under_Rule _312-A.NA.pdf	218AFBC75526D92A2558CE49534358F8865C4FC792FFE2B82 55B94734F8214D6437F2A04C7452DC61D764A1E653942A28D EBE5293D83772B00E4FA0B9765E78D
2025-03-05_MST- 0561_410864-US- CNT_Amendment_Under_Rule _312-CLM.pdf	214C49E9062700D83E443E46E50499631A34A078497103D9F0 A933D521FB4BC25FB7F9B4B8916033B9BCA9DFD2BE64D9F4 9310DA73EE7AA50FB761708337B807
2025-03-05_MST- 0561_410864-US- CNT_Amendment_Under_Rule _312-REM.pdf	9F4F03132698931E39B8B4BDA6C75FEE9494DBBAA1FA1EB7 994CDD7FD0C23D8085AD830E255288E629A9C12CFD31FBE3 6EF3D92A8DF8A20487F30214E0EE327B

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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NOTICE OF ALLOWANCE AND FEE(S) DUE

141674 7590 01/17/2025
Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Suite 1100
Washington, DC 20005

EXAMINER	
WEHOVZ, OSCAR	
ART UNIT	PAPER NUMBER
2161	

DATE MAILED: 01/17/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414

TITLE OF INVENTION: DOCUMENT RE-FINDING SYSTEM

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	04/17/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414
TITLE OF INVENTION: DOCUMENT RE-FINDING SYSTEM				

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	04/17/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
WEHOVZ, OSCAR	2161	707-722000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864-	4414
			US CNT	
			EXAMINER	
			WEHOVZ, OSCAR	
			ART UNIT	PAPER NUMBER
			2161	
DATE MAILED: 01/17/2025				

141674 7590 01/17/2025
Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
Suite 1100
Washington, DC 20005

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 18/541,216	Applicant(s) RAMOS et al.	
	Examiner OSCAR WEHOVZ	Art Unit 2161	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to December 9, 2024.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in **ABANDONMENT** of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ **CORRECTED DRAWINGS** (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ **DEPOSIT OF and/or INFORMATION** about the deposit of **BIOLOGICAL MATERIAL** must be submitted. Note the attached Examiner's comment regarding **REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL**.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☒ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/OSCAR WEHOVZ/ Examiner, Art Unit 2161	/ETIENNE P LEROUX/ Primary Examiner, Art Unit 2161
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DETAILED ACTION

1. This action is responsive to Applicant amendments filed on December 9, 2024.
2. Preliminary amendments filed on December 9, 2024 have been acknowledged and considered.
3. The terminal disclaimer filed on December 30, 2024 has been acknowledged and considered.
4. Claims 1-20 are allowed.

Notice of Pre-AIA or AIA Status

5. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

6. Applicant's Remarks, filed November 18, 2024, has been fully considered and entered.

Accordingly, Claims 1-20 are pending in this application. Claims 1-20 have been amended. Claims 1, 13 and 19 are independent claims. In light of Applicant's Amendments, the non-statutory double patenting rejection of claims 1-12 and 19-20, the statutory double patenting rejection of claims 13-18 and the 101 rejection of claims 19-20 have been withdrawn.

REASONS FOR ALLOWANCE

7. The following is an examiner's statement of reasons for allowance:

After further searches and consideration the Examiner concluded that the claimed invention has been recited in such a manner that the independent claim is not thought by any prior art of record. Independent claims 1, 13 and 19 are considered allowable since when reading the claim in light of the specification, none of the prior art of record either alone or in combination fairly disclose or suggest the combination of limitations specified in the independent claims, including at least:

In claims 1:

“generating, for each concept clip of a set of concept clips, an embedding, wherein the set of concept clips corresponds to a concept searched by a user; generating a concept embedding based on the embeddings of the set of concept clips; determining a semantic relationship between the concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and based on the determined semantic relationship, providing, for display at a client device, an indication of the semantic relationship between the concept and one or more document clips of the one or more document embeddings.”

In claims 13:

“generate, for each concept clip of a set of concept clips that correspond to a concept searched by a user using a machine learning model, an embedding; generate a concept embedding based on the embeddings of the set of concept clips; determine a semantic relationship between the concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and generate a graphical user interface based on the determined semantic relationship depicting the semantic relationship between the concept and one or more document clips of the one or more document embeddings.”

In claims 19:

“generating, for each concept clip of a set of concept clips, an embedding, wherein the set of concept clips corresponds to a concept searched by a user; generating a concept embedding based on the embeddings of the set of concept clips; determining a semantic relationship between the concept embedding and one or more document embeddings that each

correspond to a document clip of a set of documents; and based on the determined semantic relationship, providing, for display at a client device, an indication of the semantic relationship between the concept and one or more document clips of the one or more document embeddings.”

8. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure:

- 1) **Schechter, History answer for re-finding search results (US Patent Application Publication No. US 20100153428 A1)**
- 2) **Birch, Browser-based navigation suggestions for task completion (US Patent Application Publication No. US 20200042567 A1)**
- 3) **Sadahiro, Unsupervised clustering of browser history using web navigational activities (US Patent Application Publication No. US 20200380051 A1)**

A summary of each reference is provided followed by an explanation as to why the reference does not teach, either solely, or in combination with the other cited references.

Schechter set forth providing a personalized history answer to a user search query based on an inference that the user is attempting to re-find information that the user has previously seen. During subsequent searches, a history answer service attempts to determine whether the user is attempting to re-find information that the user has previously seen. If the query appears to be an attempt to access previously seen information, a personalized history

answer is displayed, which is a representation of one or more search results that the user may be attempting to re-find.

However, Schechter does not teach, either solely, or in combination with the other cited references, none of the above limitations for independent claims 1, 8 and 15.

Birch set forth acquiring data about a plurality of web pages rendered within one or more browser tabs of a web browser executing on at least one computing device and processing the acquired data and grouping web pages that are determined to be topically-related to content displayed in the web browser and provide navigation suggestions, which provides the web pages that the user is likely to re-visit.

However, Birch does not teach, either solely, or in combination with the other cited references, none of the above limitations for independent claims 1, 8 and 15.

Sadahiro set forth clustering of browser history using web navigational activities, determining a correlation score between web pages in order to group and rank them into a cluster of previously visited web pages, identifying a cluster of the clusters having a greatest ranking and providing, to the user, an indication of a web page of the identified cluster as a suggested web page to revisit.

However, Sadahiro does not teach, either solely, or in combination with the other cited references, none of the above limitations for independent claims 1, 8 and 15.

9. Based on the prior arts of record, Examiner concluded that the claimed invention as set forth in the independent claim in the subject application is allowable. Any comments

considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to OSCAR WEHOVZ whose telephone number is (571)272-3362. The examiner can normally be reached 8:00am - 5:00pm ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, APU M MOFIZ can be reached on (571) 272-4080. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Application/Control Number: 18/541,216
Art Unit: 2161

Page 7

/OSCAR WEHOVZ/
Examiner, Art Unit 2161

/ETIENNE P LEROUX/
Primary Examiner of Art Unit 2161

<i>Examiner-Initiated Interview Summary</i>	Application No. 18/541,216	Applicant(s) RAMOS et al.		
	Examiner OSCAR WEHOVZ	Art Unit 2161	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
OSCAR WEHOVZ	Examiner	Telephonic
ETHAN BELL	Attorney of Record	

Date of Interview: 30 December 2024

Issues Discussed:

Non-statutory Double Patenting

The Examiner contacted Attorney Ethan Bell (Reg. No. 75,549) to suggest filing a Terminal Disclaimer in order to expedite prosecution and avoid a Non-Statutory Double Patenting rejection in view of Patent 11847178. Attorney filed the Terminal Disclaimer.

/OSCAR WEHOVZ/ Examiner, Art Unit 2161	/ETIENNE P LEROUX/ Primary Examiner, Art Unit 2161
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Notice of References Cited	Application/Control No. 18/541,216		Applicant(s)/Patent Under Reexamination RAMOS et al.	
	Examiner OSCAR WEHOVZ		Art Unit 2161	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20100153428-A1	06-2010	SCHECHTER; MICHAEL AUSTIN	G06F16/9535	707/E17.014
*	B	US-20200042567-A1	02-2020	Birch; Samuel	G06F40/134	1/1
*	C	US-20200380051-A1	12-2020	Sadahiro; Ken M.	G06F16/9537	1/1
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Issue Classification</i> 	Application/Control No. 18/541,216	Applicant(s)/Patent Under Reexamination RAMOS et al.
	Examiner OSCAR WEHOVZ	Art Unit 2161

CPC						
Symbol					Type	Version
G06F	/	16	/	9535	F	2019-01-01
G06F	/	16	/	9538	I	2019-01-01

CPC Combination Sets							
Symbol				Type	Set	Ranking	Version
	/		/				

/OSCAR WEHOVZ/ Examiner, Art Unit 2161 (Assistant Examiner)		31 December 2024 (Date)	Total Claims Allowed: 20	
/ETIENNE P LEROUX/ Primary Examiner, Art Unit 2161 (Primary Examiner)		31 December 2024 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/541,216	Applicant(s)/Patent Under Reexamination RAMOS et al.
	Examiner OSCAR WEHOVZ	Art Unit 2161

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F16/9535	/	16	/ 9535
G06F16/9538	/	16	/ 9538
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

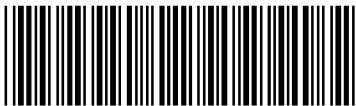
CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

/OSCAR WEHOVZ/ Examiner, Art Unit 2161 (Assistant Examiner)	31 December 2024 (Date)	Total Claims Allowed: 20	
/ETIENNE P LEROUX/ Primary Examiner, Art Unit 2161 (Primary Examiner)	31 December 2024 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/541,216	Applicant(s)/Patent Under Reexamination RAMOS et al.
	Examiner OSCAR WEHOVZ	Art Unit 2161

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	19	19										
2	2	11	11	20	20										
3	3	12	12												
4	4	13	13												
5	5	14	14												
6	6	15	15												
7	7	16	16												
8	8	17	17												
9	9	18	18												

/OSCAR WEHOVZ/ Examiner, Art Unit 2161 (Assistant Examiner)	31 December 2024 (Date)	Total Claims Allowed: 20	
/ETIENNE P LEROUX/ Primary Examiner, Art Unit 2161 (Primary Examiner)	31 December 2024 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Search Notes</i> 	Application/Control No. 18/541,216	Applicant(s)/Patent Under Reexamination RAMOS et al.
	Examiner OSCAR WEHOVZ	Art Unit 2161

CPC - Searched*		
Symbol	Date	Examiner
(G06F16/9535 or G06F16/9538 or G06F16/93 or G06F16/9038).cpc.	12/31/2024	OW

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search performed in PALM/DAV and PE2E Search	12/31/2024	OW
Assignee search performed in PALM/DAV and PE2E Search	12/31/2024	OW
IP_com/google prior art search	12/31/2024	OW
PE2E Search Limited classification and text searches - see attached search history	12/31/2024	OW

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	US-PGPUB, USPAT interference text search - see attached search history	12/31/2024	OW

/OSCAR WEHOVZ/ Examiner, Art Unit 2161	
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Bibliographic Data

Application No: 18/541,216

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /OSCAR WEHOVZ/

Examiner's Signature

Initials

Title: DOCUMENT RE-FINDING SYSTEM

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/15/2023	707	2161	
RULE			

APPLICANTS

Microsoft Technology Licensing, LLC, Redmond, WA, UNITED STATES

INVENTORS

Gonzalo A. RAMOS, Kirkland, WA, UNITED STATES

Jin A. SUH, Seattle, WA, UNITED STATES

Shiqian Rachel NG, Seattle, WA, UNITED STATES

Christopher Alan MEEK, Kirkland, WA, UNITED STATES

Haekyu PARK, Atlanta, GA, UNITED STATES

CONTINUING DATA

This application is a CON of 17683595 03/01/2022 PAT 11847178

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

12/28/2023

STATE OR COUNTRY

UNITED STATES

ADDRESS

FILING FEE RECEIVED

\$1,820

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	42233	((("MICROSOFT") near3 ("TECHNOLOGY") near3 ("LICENSING") near3 ("LLC"))).AS,AANM.	(US-PGPUB; USPAT)	OR	ON	ON	2024/07/01 03:14 PM
L2	617	((("RAMOS") near3 ("Gonzalo")) OR (("SUH") near3 ("Jin")) OR (("NG") near3 ("Shiqian")) OR (("MEEK") near3 ("Christopher")) OR (("PARK") near3 ("Haekyu"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:15 PM
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L4	2880	L3 AND (L2 OR L1)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:15 PM
L5	66	L4 AND (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:17 PM
L6	83	L3 AND (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3) AND ((embedding OR nodes) WITH (concept OR semantic\$4))	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:18 PM
L7	151	(find\$3 adj7 previous\$2) AND paragraph AND ((embedding OR nodes) WITH (concept OR semantic\$4))	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:19 PM
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L9	403	L8 AND L3	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:20 PM
L10	91	L9 AND embedding	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:20 PM

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L23	66	L15 AND (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3)	SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/12/31 07:40 AM
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L26	74	L25 AND relationship AND documents	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/12/31 07:43 AM
L27	49	L26 AND ((semantic OR relationship) WITH concept)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/12/31 07:45 AM

PE2E SEARCH - Search History (Interference)

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N7	73	L25 AND relationship AND documents	(US-PGPUB; USPAT)	OR	ON	ON	2024/12/31 07:48 AM



Search History

Report Run Date: 2024-12-31 UTC
Searches: 6
Markers: 0

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5	2024-12-31 12:52 UTC	112,087	4
6	2024-12-31 12:52 UTC	54,164	5



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Luxembourg Applications | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Patents | Israel Applications | Jordan Patents | Jordan Applications | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Saudi Arabia Applications | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Patents | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: SMPTE Periodicals | IEEE Xplore Publications: SMPTE Conferences | IEEE Xplore Publications: SMPTE Standards | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IEEE Xplore Publications: Ericsson Technology Review | IEEE Xplore Publications: Manning Books | IEEE Xplore Publications: Packt Books | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2024-12-31 12:51 UTC | 26,966,219 results from

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Document

Concept Modifiers: None

Filters: None

2 2024-12-31 12:51 UTC | 1,250,175 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 1](#)

Concept Modifiers (1):

[1. More Like Text:](#) embedding

Filters: None

3 2024-12-31 12:52 UTC | 357,737 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 1](#)

Concept Modifiers (2):

[2. More Like Text:](#) concept

[1. More Like Text:](#) embedding

Filters: None

4 2024-12-31 12:52 UTC | 112,087 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 1](#)

Concept Modifiers (3):

[3. More Like Text:](#) semantic

[2. More Like Text:](#) concept

[1. More Like Text:](#) embedding

Filters: None

5 2024-12-31 12:52 UTC | 112,087 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 1](#)

Concept Modifiers (3):

[3. More Like Text:](#) semantic relationship

[2. More Like Text:](#) concept

[1. More Like Text:](#) embedding

Filters: None

6 2024-12-31 12:52 UTC | 54,164 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Same as Search 1

Concept Modifiers (4):

- 4. More Like Text: re-finding
- 3. More Like Text: semantic relationship
- 2. More Like Text: concept
- 1. More Like Text: embedding

Filters: None

Web Search History

date, time	web site	search string
12/31/2024 7:49:23 AM	Google Patents	Search Term 1 (document re-finding)
12/31/2024 7:49:44 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding
12/31/2024 7:49:50 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding Search Term 3 concept
12/31/2024 7:49:52 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding Search Term 3 concept Search Term 4 s
12/31/2024 7:49:55 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding Search Term 3 concept Search Term 4 semantic
12/31/2024 7:50:02 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding Search Term 3 concept Search Term 4 semantic Search Term 5 relationship

date, time	web site	search string
12/31/2024 7:50:16 AM	Google Patents	Search Term 1 (document re-finding) Search Term 2 embedding Search Term 3 concept Search Term 4 semantic Search Term 5 relationship Search Term 6 (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3)
12/31/2024 7:50:22 AM	Google Scholar	(document re-finding) embedding concept semantic relationship (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3)



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TERMINAL DISCLAIMER TO OBVIATE A DOUBLE PATENTING REJECTION OVER A PRIOR PATENT

APPLICATION #	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET #
18541216	12/15/2023	Gonzalo RAMOS	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM



Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action



This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.

Owner	Percent interest
Microsoft Technology Licensing, LLC	100%
Total	100%

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Application #	Filing Date
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Patent #

11847178

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Terminal disclaimer fee under 37 CFR 1.20(d) included with Electronic Terminal Disclaimer request.

Applicant claims the following entity status:

Regular Undiscounted

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I certify, in accordance with 37 CFR 1.4(d)(4) that I am: An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application

Signature	Name	Registration #
/Ethan W. Bell/	Ethan Bell	75549

* Statement under 37 CFR 3.73(b) is required if terminal disclaimer is signed by the assignee (owner). Form PTO/SB/96 may be used for making this certification. See MPEP 324.



UNITED STATES
PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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APPROVAL LETTER

APPLICATION #
18/541,216

FILING DATE
12/15/2023

APPLICANT/PATENT UNDER REEXAMINATION
Gonzalo RAMOS

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Electronic terminal disclaimer filed on 12/30/2024



Approved

This patent is subject to a Terminal Disclaimer

Approved / Disapproved by: Electronic Terminal Disclaimer automatically approved

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	12/30/2024 03:17:13 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	68553336	FILING DATE	12/15/2023
CUSTOMER #	141674	FIRST NAMED INVENTOR	Gonzalo A. RAMOS
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Ethan Bell

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
petition-request.pdf	3	Terminal Disclaimer-Filed (Electronic)	44 KB
grantLetter.pdf	1	Terminal Disclaimer-Electronic-Approved	18 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
petition-request.pdf	BAD815A112097DBECEDF3BA50DD1435012184CF82017C9D0 AB321F4F16FDD0B0474D4B7978A2F45E573ECFC3F7B32205B D722712C28F5495219A84B1B12C6514
grantLetter.pdf	E1D09CA7E75173E87EEAE61FAB7A8319CCF45347ECDD0B32

91807DE2AA4C82487C566B6B2E0416ED70CFF506D14552443
E33931BFD8A0D94944544ED2AD3D96C

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	12/30/2024 03:17:13 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	68553336	AUTHORIZED BY	Ethan Bell
CUSTOMER #	141674	FILING DATE	12/15/2023
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Gonzalo A. RAMOS

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
DA / 500573	E2024BTF17296508	Ethan Bell

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1814	STATUTORY DISCLAIMER, INCLUDING TERMINAL DISCLAIMER	170.00	1	170.00
TOTAL AMOUNT:				\$170.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C.

371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional) MST-0561/410864-US-CNT
Application Number 18/541,216	Filed December 15, 2023	
For DOCUMENT RE-FINDING SYSTEM		
Art Unit 2161	Examiner Oscar Wehovz	

This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application.

The requested extension and fee are as follows (check time period desired and enter the appropriate fee below):

	<u>Fee</u>	<u>Small Entity Fee</u>	<u>Micro Entity Fee</u>	
☐ One month (37 CFR 1.17(a)(1))	\$220	\$88	\$44	\$ _____
☒ Two months (37 CFR 1.17(a)(2))	\$640	\$256	\$128	\$ <u>640</u>
☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$592	\$296	\$ _____
☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$928	\$464	\$ _____
☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,264	\$632	\$ _____

☐ Applicant asserts small entity status. See 37 CFR 1.27.☐ Applicant certifies micro entity status. See 37 CFR 1.29.
Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously.☐ A check in the amount of the fee is enclosed.☐ Payment by credit card. Form PTO-2038 is attached.☐ The Director has already been authorized to charge fees in this application to a Deposit Account.☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to
Deposit Account Number 50-0573.☒ Payment made via USPTO patent electronic filing system.**WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.**

I am the

☐ applicant.☒ attorney or agent of record. Registration number 75,549.☐ attorney or agent acting under 37 CFR 1.34. Registration number _____./Ethan W. Bell/

Signature

December 9, 2024

Date

Ethan W. Bell

Typed or printed name

303.927.5220

Telephone Number

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.☐ * Total of _____ forms are submitted.

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- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:

Gonzalo A. RAMOS et al.

Confirmation No.: 4414

Application No. 18/541,216

Art Unit: 2161

Filed: December 15, 2023

Examiner: Oscar Wehovz

For: DOCUMENT RE-FINDING SYSTEM

RESPONSE TO OFFICE ACTION OF JULY 8, 2024

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

Dear Commissioner:

In response to the Office Action of July 8, 2024, please enter the following amendments and consider the following remarks:

Amendments to the Claims are reflected in the Listing of Claims which begins on page 2 of this paper.

Remarks begin on page 9 of this paper.

AMENDMENTS TO THE CLAIMS

This listing of claims will replace all prior versions and listings of claims in the application. Please amend the claims as indicated below.

Listing of Claims:

1. (Currently Amended) A method for assisting users in re-finding documents, the method comprising:
~~generating embeddings for document clips related to respective documents among a plurality of documents;~~
~~receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;~~
~~generating, for each concept clip of a set of concept clips, an embedding embeddings for respective concept clips in the first, wherein the set of concept clips corresponds to a concept searched by a user;~~
~~generating a [[first]] concept embedding based on a combination of the embeddings of the set of generated for the respective concept clips; in the first set of concept clips;~~
~~determining a semantic relationship relationships between the [[first]] concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;~~
~~based on the determined semantic relationship, providing, for display at a client device, an indication of generating a graphical user interface depicting the semantic relationship relationships between the [[first]] concept and [[the]] one or more document clips of the one or more document embeddings; and~~
~~causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.~~
2. (Currently Amended) The method of claim 1, wherein the ~~plurality~~ set of documents comprises a ~~plurality of web pages~~ web page in a browsing history of the user.
3. (Currently Amended) The method of claim 1, wherein the document clip includes ~~clips related to content of a particular document among the plurality of documents include~~

~~one or more of i) one or more a text paragraph paragraphs~~ parsed out from ~~the particular a~~ document of the set of documents or an image from the document. and ii) ~~one or more images~~ from ~~the particular document.~~

4. (Currently Amended) The method of claim 1, wherein the document clip includes ~~clips related to content of a particular document include~~ at least one user-defined clip that includes information not extracted from ~~[[the]]~~ a document of the one or more documents.

5. (Currently Amended) The method of claim 1, wherein ~~the first set of clips defining the first concept~~ the set of concept clips includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user.

6. (Currently Amended) The method of claim 1, wherein ~~generating respective embeddings for the document clips comprises generating the~~ document embeddings for the document clips were generated using a machine learning model ~~configured to generate embeddings~~ based on content, and wherein the document embeddings capture semantic meaning semantics of the content, and wherein ~~generating the~~ embeddings for respective concept clips ~~comprises generating the embeddings for the respective concept clips~~ are generated using the machine learning model.

7. (Currently Amended) The method of claim 1, wherein the concept embedding is generated ~~generating the first concept embedding comprises generating the first concept embedding~~ based on a weighted combination of the embeddings generated for the ~~respective concept clips in the first set of concept clips.~~

8. (Currently Amended) The method of claim 1, wherein the indication of the semantic relationship comprises generating the graphical user interface includes placing graphical representations of the one or more document clips in a multidimensional canvas, wherein the graphical representations of the document clips are placed relative to each other based on calculated distances between the one or more document embeddings ~~generated~~ for the corresponding document clips.

9. (Currently Amended) The method of claim 8, wherein providing the indication of the semantic relationship generating the graphical user interface further comprises:

placing a graphical representation of the [[first]] concept at a first location in the multidimensional canvas;

determining locations of the graphical representations of the document clips relative to the first location of the graphical representation of the first concept based on calculated distances between the document embeddings ~~generated for the corresponding document clips~~ and the [[first]] concept embedding; and

positioning the graphical representations of the document clips in relation to the graphical representation of the [[first]] concept in the multidimensional canvas in accordance to the determined distances.

10. (Currently Amended) The method of claim 9, wherein the set of concept clips is a first set of concept clips and the concept is a first concept, and the method further comprises:
~~comprising:~~

receiving a second set of concept clips defining a second concept for searching for content of interest to the user; ~~in the plurality of documents;~~

placing a graphical representation of the second concept at a second location in the multidimensional canvas; and

updating locations of the graphical representations of the document clips such that:

i) graphical representations of document clips that are more semantically similar to the first concept than the second concept are placed closer to the graphical representation of the first concept in the multidimensional canvas; and

ii) graphical representations of document clips that are more semantically similar to the second concept than the first concept are placed closer to the graphical representation of the second concept in the multidimensional canvas.

11. (Currently Amended) The method of claim 10, wherein updating locations of the graphical representations of the document clips comprises:

generating embeddings for respective concept clips in the second set of concept clips;

generating a second concept embedding based ~~on a combination of~~ the embeddings generated for the respective concept clips in the second set of concept clips;

determining updated locations of the graphical representations of the document clips in the multidimensional canvas based on:

- i) the first location of the graphical representation of the first concept in the multidimensional canvas,
- ii) calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding,
- iii) the second location of the graphical representation of the second concept in the multidimensional canvas, and
- iv) calculated distances between the embeddings generated for the corresponding document clips and the second concept embedding; and moving the graphical representations of the document clips to the updated locations in the multidimensional canvas.

12. (Currently Amended) The method of claim 10, further comprising:
detecting that the graphical representation of the first concept has been repositioned to a new location in the multidimensional canvas by the user;[[,]] and
in response to detecting that the graphical representation of the first concept has been repositioned, repositioning graphical representations of the document clips in the multidimensional canvas such that graphical representations of document clips that are more semantically similar to the first concept than the second concept move closer to the new location of the graphical representation of the first concept.

13. (Currently Amended) A system for assisting users in re-finding documents, the system comprising:
one or more computer readable storage media; and
program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to:
~~generate, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents;~~
~~receive a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;~~

generate, for each concept clip of a set of concept clips that correspond to a concept searched by a user using [[the]] a machine learning model, an embedding;
~~embeddings for respective concept clips in the first set of concept clips;~~

generate a [[first]] concept embedding based on ~~a combination of the~~
embeddings of the set of ~~generated for the respective concept clips in the first set of~~
concept clips;

determine ~~a semantic relationship~~ relationships between the [[first]] concept
embedding and one or more document embeddings that each correspond to a
document clip of a set of documents; and the document clips based on i) the
embeddings generated for the document clips and ii) the concept embedding;

generate a graphical user interface based on the determined semantic
relationship depicting the semantic relationship ~~relationships~~ between the [[first]]
concept and [[the]] one or more document clips of the one or more document
embeddings; ~~and~~

~~cause display of the graphical user interface to be rendered at a client device;~~
~~wherein the graphical user interface is operable to enable re-finding a document;~~
~~among the plurality of documents, having the content of interest to the user.~~

14. (Currently Amended) The system of claim 13, wherein the ~~plurality~~ set of documents
comprises a ~~plurality of web pages~~ a web page in a browsing history of the user.

15. (Currently Amended) The system of claim 13, wherein the document clip includes
~~clips related to content of a particular document among the plurality of documents include~~
~~one or more of i) one or more a text paragraph~~ paragraphs ~~parsed out from the particular a~~
document of the set of documents, an image from the document, or a user-defined clip that
includes information not extracted from the document. ii) one or more images from the
~~particular document and iii) at least one user-defined clip that includes information not~~
~~extracted from the document.~~

16. (Currently Amended) The system of claim 13, wherein ~~the first set of clips defining~~
~~the first concept~~ the set of concept clips includes one or more of i) one or more custom text
paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more
documents by the user and iii) one or more images provided by the user.

17. (Currently Amended) The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the [[first]] concept embedding based on a weighted combination of the embeddings generated for the ~~respective concept clips in the first~~ set of concept clips.

18. (Currently Amended) The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the graphical user interface at least by:

placing graphical representations of the document clips in a multidimensional canvas in the graphical user interface, wherein the graphical representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips;

placing a graphical representation of the [[first]] concept at a [[first]] location in the multidimensional canvas;

determining locations of the graphical representations of the document clips relative to the [[first]] location of the graphical representation of the [[first]] concept based on calculated distances between the embeddings generated for the corresponding document clips and the [[first]] concept embedding; and

positioning the graphical representations of the document clips in relation to the graphical representation of the [[first]] concept in the multidimensional canvas in accordance to the determined distances.

19. (Currently Amended) A computer storage medium comprising computer executable instructions, which when executed by at least one processor, perform a method comprising:

~~generating embeddings for document clips related to respective documents among a plurality of documents;~~

~~receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;~~

generating, for each concept clip of a set of concept clips, an embedding ~~embeddings~~ for ~~respective concept clips in the first~~, wherein the set of concept clips corresponds to a concept searched by a user;

generating a [[first]] concept embedding based on ~~a combination of the embeddings of the set of generated for the respective concept clips in the first set of concept clips;~~

determining a semantic relationship ~~relationships~~ between the [[first]] concept embedding and one or more document embeddings that each correspond to a document clip of a set of documents; and ~~the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;~~

based on the determined semantic relationship, providing, for display at a client device, an indication of ~~generating a graphical user interface depicting the semantic relationship~~ relationships between the [[first]] concept and [[the]] one or more document clips of the one or more document embeddings; ~~and~~

~~causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.~~

20. (Currently Amended) The computer storage medium of claim 19, wherein the method further comprises:

~~placing representations of the document clips in a multidimensional canvas in the graphical user interface, wherein the representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips;~~

placing a representation of the [[first]] concept at a [[first]] location in the multidimensional canvas;

determining locations of [[the]] representations of the document clips relative to the [[first]] location of the representation of the [[first]] concept based on calculated distances between the document embeddings generated for the corresponding document clips and the [[first]] concept embedding; and

placing ~~positioning~~ the representations of the document clips in relation to the representation of the [[first]] concept in the multidimensional canvas in accordance to the determined distances.

REMARKS

The following remarks fully respond to the Office Action of July 08, 2024 (“Office Action”). By the present Amendment, claims 1-20 have been amended, no claims have been cancelled, and no claims have been added. Thus, claims 1-20 remain present for examination.

Double Patenting Rejections

Claims 1-12 and 19-20 were rejected under the doctrine of obviousness-type double patenting, over claims 1-12 and 19-20 of U.S. Patent No. 11,847,178. Applicant respectfully requests that the double patenting issue be held in abeyance until agreement on the scope of allowable claims is reached. When/if the only remaining rejections to otherwise allowable claims are double patenting rejections, Applicants may submit a terminal disclaimer if appropriate. The Examiner is respectfully encouraged to evaluate the double patenting issue as claims are amended and renew these rejections, if proper.

Additionally, claims 13-18 were rejected under 35 U.S.C. § 101 and statutory double patenting as claiming the same invention as that of claims 13-18 of U.S. Patent No. 11,847,178. Claims 13-18 are amended herein to address the statutory double patenting rejection, thereby rendering the rejection moot. Withdrawal of the rejection is therefore respectfully requested.

Rejections under 35 U.S.C. §101

Claims 19-20 were rejected under 35 USC 101 because the claimed invention is directed to non-statutory subject matter. In particular, the Office Action asserts the claims are “drawn to a computer storage medium,” which “typically covers from form of non-transitory tangible media and transitory propagating signals.” Office Action, pg. 2. Additionally, the Office Action asserts “the specification is silent to limit that such medium exclude signals.” Office Action, pg. 2.

Applicant respectfully disagrees, as at least paragraph [0069] of the present application notes “computer storage media does not include a carrier wave or other propagated or modulated data signal,” such that the “computer storage medium” of claims 19-20 cannot be interpreted to include a propagating signal per se. Withdrawal of the § 101 rejection is therefore respectfully requested.

CONCLUSION

This Amendment fully responds to the Office Action. Still, there may be arguments and rejections that are not directly addressed herein due to the fact they are rendered moot in light of the preceding arguments in favor of patentability. Hence, failure to directly address an argument raised in the Office Action should not be taken as an indication that Applicant believes the argument has merit. Furthermore, the claims of the present application may include other elements, not discussed herein, which are not shown, taught, or otherwise suggested by the references of record. Accordingly, the preceding arguments in favor of patentability are advanced without prejudice to other bases of patentability.

This Amendment is filed concurrently with a Petition for Extension of Time. It is believed that no further fees are due at this time. However, the Commissioner is hereby authorized to charge any deficiencies or credit any overpayment with respect to this patent application to deposit account number 50-0573.

For the foregoing reasons, reconsideration and allowance is respectfully requested. The Examiner is invited to call the undersigned if it would be helpful to advance prosecution of this application.

Respectfully submitted,

Date: December 9, 2024

/Ethan W. Bell/
Ethan W. Bell, Reg. No. 75,549



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	12/09/2024 03:14:03 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	68289014	FILING DATE	12/15/2023
CUSTOMER #	141674	FIRST NAMED INVENTOR	Gonzalo A. RAMOS
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Ethan Bell

Documents

TOTAL DOCUMENTS: 4

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2024-12-09_MST-0561_410864-US-CNT_Response_to_Office_Action.pdf	10	-	127 KB
2024-12-09_MST-0561_410864-US-CNT_Response_to_Office_Action-A....pdf (1-1)	1	Amendment/Request for Reconsideration-After Non-Final Rejection	81 KB
2024-12-09_MST-0561_410864-US-CNT_Response_to_Office_Action-CLM.pdf (2-8)	7	Claims	102 KB
2024-12-09_MST-0561_410864-US- (9-10)	2	Applicant Arguments/Remarks Made in an Amendment	99 KB

CNT_Response_to_Office_
Action-REM.pdf

2024-12-09_MST-
0561_410864-US-
CNT_EOT.pdf

2

Extension of Time

284 KB

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DOCUMENT

MESSAGE DIGEST(SHA-512)

2024-12-09_MST-
0561_410864-US-
CNT_Response_to_Office_Acti
on.pdf

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on-CLM.pdf

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2024-12-09_MST-
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CNT_EOT.pdf

79A60934906C73C8D2D1B25FFAFFD4FEE5064BF04DE35696E
7236844AD7CD0BEB9E28CEBBD852F6C8E4CB10B123E313F3
CFD719AFC38EBEB09604FEF2CE906EA

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the

International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/541,216	12/09/2024 03:14:03 PM Z ET	MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4414	FILED BY	Irina Mikitiouk
PATENT CENTER #	68289014	AUTHORIZED BY	Ethan Bell
CUSTOMER #	141674	FILING DATE	12/15/2023
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Gonzalo A. RAMOS

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 5984	E2024B9F14555255	Irina Mikitiouk
PRE-AUTHORIZED ACCOUNT	PRE-AUTHORIZED CATEGORY	
500573	37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.21 (Miscellaneous fees and charges)	

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
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TOTAL AMOUNT:				\$640.00

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 18/541,216	Filing Date 12/15/2023	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)	
AMENDMENT	12/09/2024	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =	0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$480 =	0	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE	0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =		
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =		
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						SLIE		
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/STEPHEN TYLER HOLLAND/		
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".								
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.								

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864- US-CNT	4414
141674	7590	07/08/2024	EXAMINER	
Faegre Drinker Biddle & Reath LLP (Microsoft)			WEHOVZ, OSCAR	
1500 K Street, N.W.			ART UNIT	
Suite 1100			PAPER NUMBER	
Washington, DC 20005			2161	
			NOTIFICATION DATE	DELIVERY MODE
			07/08/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentdocketingdvr@faegredrinker.com
usdocket1@microsoft.com

DETAILED ACTION

1. This action is responsive to application filed on December 15, 2023.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 101

3. 35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

4. Claims 19-20 are rejected under 35 USC 101 because the claimed invention is directed to non-statutory subject matter.

Referring to claim 19, the broadest reasonable interpretation of a claim drawn to a computer storage medium comprising computer executable instructions (also called machine readable medium and other such variations) typically covers from form of non-transitory tangible media and transitory propagating signals. The specification is silent to limit that such medium exclude signals. Therefore covers transitory propagating signal per se (non-statutory subject matter). See MPEP 2111.01. (When the broadest reasonable interpretation of a claim covers a signal per se, the claim must be rejected under 35 USC 101 as covering non-statutory subject matter. See *In re Nuijten*, 500 F.3d 1346, 1356-57 (Fed. Cir. 2007) (transitory embodiments are not directed to statutory subject matter) and Interim Examination Instructions for Evaluating Subject Matter Eligibility Under 35 USC 101, Aug. 24, 2009; p.2.) OG Notice - Jan. 28, 2010. It is respectfully suggested to amend the

term in a way to preclude an interpretation of "propagating signal per se", such as "A non-transitory computer storage medium".

Therefore, the claims are non-statutory. Per claim 20, it does not cure the deficiency in claim 19 and is also rejected based on its dependency on claim 19.

Double Patenting

5. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for

applications not subject to examination under the first inventor to file provisions of the AIA. A

terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

6. Claims 1-12 and 19-20 are rejected on the ground of nonstatutory double patenting as being unpatentable over the claims 1-12 and 19-20 of prior U.S. Patent No. 11,847,178. Although the claims at issue are not identical, they are not patentably distinct from each other because they are directed to substantially the same invention.

Instant application	U.S. Patent No. 11,847,178
Claim 1. A method for assisting users in re-finding documents, the method comprising: generating embeddings for document clips related to respective documents among a plurality of documents;	Claim 1. A method for assisting users in re-finding documents, the method comprising: generating, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents;

receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generating embeddings for respective concept clips in the first set of concept clips; generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips; determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding; generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.	receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generating embeddings for respective concept clips in the first set of concept clips; generating, using the machine learning model, a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips; determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding; generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.
Claim 2	Claim 2
Claim 3	Claim 3
Claim 4	Claim 4
Claim 5	Claim 5
Claim 6	Claim 6
Claim 7	Claim 7
Claim 8	Claim 8
Claim 9	Claim 9
Claim 10	Claim 10
Claim 11	Claim 11
Claim 12	Claim 12
Claim 19. A computer storage medium comprising computer executable instructions, which when executed by at	Claim 19. A non-transitory computer storage medium comprising computer

least one processor, perform a method comprising: generating embeddings for document clips related to respective documents among a plurality of documents; receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generating embeddings for respective concept clips in the first set of concept clips; generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips; determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding; generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.	executable instructions, which when executed by at least one processor, perform a method comprising: generating, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents; receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generating, using the machine learning model, embeddings for respective concept clips in the first set of concept clips; generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips; determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding; generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.
Claim 20	Claim 20

7. A rejection based on double patenting of the “same invention” type finds its support in the language of 35 U.S.C. 101 which states that “whoever invents or discovers any new and useful process... may obtain a patent therefor...” (Emphasis added). Thus, the term “same invention,” in this context,

means an invention drawn to identical subject matter. See *Miller v. Eagle Mfg. Co.*, 151 U.S. 186 (1894); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Ockert*, 245 F.2d 467, 114 USPQ 330 (CCPA 1957).

A statutory type (35 U.S.C. 101) double patenting rejection can be overcome by canceling or amending the claims that are directed to the same invention so they are no longer coextensive in scope. The filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. 101.

8. Claims 13-18 are rejected under 35 U.S.C. 101 as claiming the same invention as that of claims 13-18 of prior U.S. Patent No. 11,847,178. This is a statutory double patenting rejection.

Instant application	U.S. Patent No. 11,847,178
Claim 13. A system for assisting users in re-finding documents, the system comprising: one or more computer readable storage media; and program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to: generate, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents; receive a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generate, using the machine learning model, embeddings for respective concept clips in the first set of concept clips; generate a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips;	Claim 13. A system for assisting users in re-finding documents, the system comprising: one or more computer readable storage media; and program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to: generate, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents; receive a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents; generate, using the machine learning model, embeddings for respective concept clips in the first set of concept clips; generate a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips;

determine semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;	determine semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;
generate a graphical user interface depicting the semantic relationships between the first concept and the document clips; and	generate a graphical user interface depicting the semantic relationships between the first concept and the document clips; and
cause display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.	cause display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.
Claim 14	Claim 14
Claim 15	Claim 15
Claim 16	Claim 16
Claim 17	Claim 17
Claim 18	Claim 18

Examiner Comments

9. For claims 1-20, no art rejection is made for these claims. Claims 1-12 and 19-20 are rejected on the grounds of nonstatutory double patenting, and Claims 13-18 are rejected on the grounds of statutory double patenting as explained above in this Office action.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to OSCAR WEHOVZ whose telephone number is (571)272-3362. The examiner can normally be reached 8:00am - 5:00pm ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, APU M MOFIZ can be reached on (571) 272-4080. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/OSCAR WEHOVZ/
Examiner, Art Unit 2161

/ETIENNE P LEROUX/
Primary Examiner of Art Unit 2161

<i>Notice of References Cited</i>	Application/Control No. 18/541,216		Applicant(s)/Patent Under Reexamination RAMOS et al.	
	Examiner OSCAR WEHOVZ		Art Unit 2161	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-11847178-B2	2023-12-19	Ramos; Gonzalo A.	G06F16/9535	1/1
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

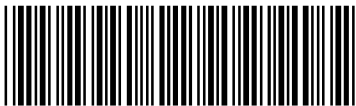
FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 18/541,216	Applicant(s)/Patent Under Reexamination RAMOS et al.
	Examiner OSCAR WEHOVZ	Art Unit 2161

CPC - Searched*		
Symbol	Date	Examiner
(G06F16/9535 or G06F16/9538 or G06F16/93 or G06F16/9038).cpc.	07/01/2024	OW

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search performed in PALM/DAV and PE2E Search	07/01/2024	OW
Assignee search performed in PALM/DAV and PE2E Search	07/01/2024	OW
IP_com/google prior art search	07/01/2024	OW
PE2E Search Limited classification and text searches - see attached search history	07/01/2024	OW

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/OSCAR WEHOVZ/ Examiner, Art Unit 2161	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	42233	((("MICROSOFT") near3 ("TECHNOLOGY") near3 ("LICENSING") near3 ("LLC"))).AS,AANM.	(US-PGPUB; USPAT)	OR	ON	ON	2024/07/01 03:14 PM
L2	617	((("RAMOS") near3 ("Gonzalo")) OR (("SUH") near3 ("Jin")) OR (("NG") near3 ("Shiqian")) OR (("MEEK") near3 ("Christopher")) OR (("PARK") near3 ("Haekyu"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:15 PM
L3	84175	(G06F16/9535 OR G06F16/9538 OR G06F16/93 OR G06F16/9038).cpc.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:15 PM
L4	2880	L3 AND (L2 OR L1)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:15 PM
L5	66	L4 AND (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:17 PM
L6	83	L3 AND (refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3) AND ((embedding OR nodes) WITH (concept OR semantic\$4))	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:18 PM
L7	151	(find\$3 adj7 previous\$2) AND paragraph AND ((embedding OR nodes) WITH (concept OR semantic\$4))	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:19 PM
L8	5730	(refind\$3 OR re-find\$3 OR re-search\$3 OR revisit\$3 OR re-visit\$3) AND semantic\$4 AND (concept OR context)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:20 PM
L9	403	L8 AND L3	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:20 PM
L10	91	L9 AND embedding	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/07/01 03:20 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 18/541,216

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /OSCAR WEHOVZ/

Examiner's Signature

Initials

Title: DOCUMENT RE-FINDING SYSTEM

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
12/15/2023	707	2161	
RULE			

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CONTINUING DATA

This application is a CON of 17683595 03/01/2022 PAT 11847178

FOREIGN APPLICATIONS

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12/28/2023

STATE OR COUNTRY

UNITED STATES

ADDRESS

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\$1,820



Search History

Report Run Date: 2024-07-01 UTC
Searches: 6
Markers: 0

Search #	Timestamp	Results	Page
1	2024-07-01 16:51 UTC	104,944,212	3
2	2024-07-01 19:27 UTC	26,048,234	4
3	2024-07-01 19:27 UTC	1,178,129	4
4	2024-07-01 19:27 UTC	324,454	4
5	2024-07-01 19:27 UTC	156,019	4
6	2024-07-01 19:28 UTC	80	5



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Luxembourg Applications | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Patents | Israel Applications | Jordan Patents | Jordan Applications | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Saudi Arabia Applications | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Patents | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: SMPTE Periodicals | IEEE Xplore Publications: SMPTE Conferences | IEEE Xplore Publications: SMPTE Standards | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IEEE Xplore Publications: Ericsson Technology Review | IEEE Xplore Publications: Manning Books | IEEE Xplore Publications: Packt Books | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2024-07-01 16:51 UTC | 104,944,212 results from

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Text

A method comprising: receiving a first event of a pre-processed event stream, the first event including a plurality of event data fields representing an occurrence of the first event; determining timing information based on the plurality of event data fields representing the occurrence of the first event; generating, based on combining the plurality of event data fields and the timing information, a second event of a discoverable event stream corresponding to the pre-processed event stream; and generating, within a time series of the discoverable event stream, based on the timing information and an event field of the plurality of event data fields, a time series entry corresponding to the second event.

Concept Modifiers: None

Filters: None

2 2024-07-01 19:27 UTC | 26,048,234 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Document

 US11847178 Document re-finding system [\(view\)](#)

Concept Modifiers: None

Filters: None

3 2024-07-01 19:27 UTC | 1,178,129 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 2](#)

Concept Modifiers (1):

[1. More Like Text:](#) embedding

Filters: None

4 2024-07-01 19:27 UTC | 324,454 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 2](#)

Concept Modifiers (2):

[2. More Like Text:](#) machine learning

[1. More Like Text:](#) embedding

Filters: None

5 2024-07-01 19:27 UTC | 156,019 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: [Same as Search 2](#)

Concept Modifiers (3):

[3. More Like Text:](#) semantic

- 2. More Like Text: machine learning
- 1. More Like Text: embedding

Filters: None

6

2024-07-01 19:28 UTC | 80 results from 

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Same as Search 2

Concept Modifiers (4):

- 4. More Like Text: re-find
- 3. More Like Text: semantic
- 2. More Like Text: machine learning
- 1. More Like Text: embedding

Filters: None

Web Search History

date, time	web site	search string
7/1/2024 3:24:06 PM	Google Patents	Search Term 1 (document re-finding)
7/1/2024 3:24:15 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page
7/1/2024 3:24:27 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page Search Term 3 previous
7/1/2024 3:24:31 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page Search Term 3 previous Search Term 4 search
7/1/2024 3:24:39 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page Search Term 3 previous Search Term 4 search Search Term 5 embedding

date, time	web site	search string
7/1/2024 3:24:49 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page Search Term 3 previous Search Term 4 search Search Term 5 embedding Search Term 6 concept
7/1/2024 3:25:16 PM	Google Patents	Search Term 1 (document re-finding) Search Term 2 web page Search Term 3 previous Search Term 4 search Search Term 5 embedding Search Term 6 concept Search Term 7 images
7/1/2024 3:25:20 PM	Google Scholar	(document re-finding) web page previous search embedding concept images



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/541,216	12/15/2023	Gonzalo A. RAMOS	MST-0561/410864-US-CNT

CONFIRMATION NO. 4414

141674

Faegre Drinker Biddle & Reath LLP (Microsoft)
1500 K Street, N.W.
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Washington, DC 20005

PUBLICATION NOTICE



OC000000070984065

Date Mailed: 04/11/2024

Title:DOCUMENT RE-FINDING SYSTEM

Publication No.US-2024-0119099-A1

Publication Date:04/11/2024

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be viewed using the USPTO's publicly available Searchable Databases via the Patent Public Search tool at www.uspto.gov. The direct link to access the Patent Public Search tool is currently <https://ppubs.uspto.gov/pubwebapp/static/pages/ppubsbasic.html>.

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Further assistance in electronically accessing the publication, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/541,216	12/15/2023		1820	MST-0561/410864-US-CNT	20	3

CONFIRMATION NO. 4414

141674

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FILING RECEIPT



0000000066368557

Date Mailed: 01/03/2024

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

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Assignment For Published Patent Application

Microsoft Technology Licensing, LLC, Redmond, WA

Power of Attorney: The patent practitioners associated with Customer Number 141674

Domestic Priority data as claimed by applicant

This application is a CON of 17/683,595 03/01/2022 PAT 11,847,178

Foreign Applications for which priority is claimed (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/28/2023

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/541,216**

Projected Publication Date: 04/11/2024

Non-Publication Request: No

Early Publication Request: No

Title

DOCUMENT RE-FINDING SYSTEM

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/541,216

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

AMENDMENT A	(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =
Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =
Application Size Fee (37 CFR 1.16(s))							OR	
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR	
				TOTAL ADD'L FEE			OR	TOTAL ADD'L FEE

AMENDMENT B	(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =
Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =
Application Size Fee (37 CFR 1.16(s))							OR	
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR	
				TOTAL ADD'L FEE			OR	TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

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A handwritten signature in black ink, reading "Kathi Vidal".

Kathi Vidal

Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

A handwritten signature in black ink, reading "Derrick Brent".

Derrick Brent

Deputy Under Secretary of Commerce for Intellectual Property and
Deputy Director of the United States Patent and Trademark Office

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Gonzalo	A.	RAMOS		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Kirkland	State/Province	WA	Country of Residenceⁱ	US
Mailing Address of Inventor:					
Address 1		c/o Microsoft Corporation			
Address 2		One Microsoft Way			
City	Redmond	State/Province	WA		
Postal Code	98052	Countryⁱ	US		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jin	A.	SUH		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Seattle	State/Province	WA	Country of Residenceⁱ	US
Mailing Address of Inventor:					
Address 1		c/o Microsoft Corporation			
Address 2		One Microsoft Way			
City	Redmond	State/Province	WA		
Postal Code	98052	Countryⁱ	US		
Inventor 3					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Shiqian	Rachel	NG		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

City	Seattle	State/Province	WA	Country of Residence ⁱ	US
------	---------	----------------	----	-----------------------------------	----

Mailing Address of Inventor:

Address 1	c/o Microsoft Corporation				
Address 2	One Microsoft Way				
City	Redmond	State/Province	WA		
Postal Code	98052	Country ⁱ	US		

Inventor 4

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Christopher	Alan	MEEK	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Kirkland	State/Province	WA	Country of Residence ⁱ	US
------	----------	----------------	----	-----------------------------------	----

Mailing Address of Inventor:

Address 1	c/o Microsoft Corporation				
Address 2	One Microsoft Way				
City	Redmond	State/Province	WA		
Postal Code	98052	Country ⁱ	US		

Inventor 5

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Haekyu		PARK	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Atlanta	State/Province	GA	Country of Residence ⁱ	US
------	---------	----------------	----	-----------------------------------	----

Mailing Address of Inventor:

Address 1	c/o Microsoft Corporation				
Address 2	One Microsoft Way				
City	Redmond	State/Province	WA		
Postal Code	98052	Country ⁱ	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		
Customer Number	141674		
Email Address	usdocket1@microsoft.com	Add Email	Remove Email

Application Information:

Title of the Invention	DOCUMENT RE-FINDING SYSTEM		
Attorney Docket Number	MST-0561/410864-US-CNT	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	12	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	141674		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation of	17683595	2022-03-01
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Remove
			Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	Microsoft Technology Licensing, LLC		
Mailing Address Information For Applicant:			
Address 1	One Microsoft Way		
Address 2			
City	Redmond	State/Province	WA
Country	US	Postal Code	98052
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

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If the Assignee or Non-Applicant Assignee is an Organization check here.



Organization Name

Microsoft Technology Licensing, LLC

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1		One Microsoft Way	
Address 2			
City	Redmond	State/Province	WA
Country ⁱ	US	Postal Code	98052
Phone Number		Fax Number	
Email Address			

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

[Add](#)**Signature:**[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application **and** either box A or B is **not** checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Ethan W. Bell/		Date (YYYY-MM-DD)	2023-12-15
First Name	Ethan W.	Last Name	Bell	Registration Number
				75549

Additional Signature may be generated within this form by selecting the Add button.

[Add](#)

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	MST-0561/410864-US-CNT
		Application Number	
Title of Invention	DOCUMENT RE-FINDING SYSTEM		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN
APPLICATION DATA SHEET (37 CFR 1.76)****Title of
Invention****DOCUMENT RE-FINDING SYSTEM**

410864-US-NP

As the below named inventor, I hereby declare that:

This declaration ☒ The attached application, or
is directed to:☐ United States application or PCT international application number _____
filed on _____.

The above-identified application was made or authorized to be made by me.

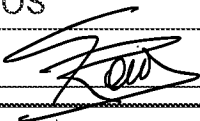
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001
by fine or imprisonment of not more than five (5) years, or both.**WARNING:**

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTORInventor: Gonzalo A. RAMOS

Date (Optional) : _____

Signature: _____


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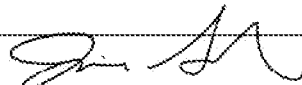
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LEGAL NAME OF INVENTOR

Inventor: Jin A. SUH

Date (Optional) : 02/26/2022

Signature: 

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LEGAL NAME OF INVENTORInventor: Shiqian Rachel NGDate (Optional) : 26 February 2022

Signature: _____

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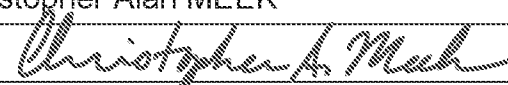
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LEGAL NAME OF INVENTORInventor: **Christopher Alan MEEK**Date (Optional) : **2/24/2022**Signature: 

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LEGAL NAME OF INVENTORInventor: Haekyu PARKDate (Optional) : Feb 24, 2022Signature: Haekyu Park

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1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DOCUMENT RE-FINDING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 17/683,595, filed March 1, 2022, now U.S. Patent No. 11,847,178, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Searching for content in various electronic documents, such as webpages, electronic files, emails, etc., is a task performed by users of various applications on a daily basis. In some cases, search for content involves re-finding documents that have been previously encountered by the user. For example, a user may wish to re-find a webpage that the user has previously explored. As another example, a user may wish to re-find an email or a document that the user has previously viewed. Typical searching systems allow the user to search for such documents by using keywords obtaining search results retrieved based on the keywords. However, re-finding documents using keywords is sometimes difficult because the user often does not sufficiently remember the content of the document to provide exact keywords for searching for the document.

[0003] It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although relatively specific problems may be discussed, it should be understood that the examples should not be limited to solving the specific problems identified in the background or elsewhere in this disclosure.

SUMMARY

[0004] In accordance with examples of the present disclosure, a re-finder system may be configured to assist users in re-finding document having content of interest to the users. In aspects, the re-finder system may allow the user to define concepts that may include clips, such as chunks of documents, images, text paragraphs, etc., that may be related to the content of interest to the user. The re-finder system may determine semantic relationships between the concepts defined by the user and clips of documents previously seen by the user, and may visually depict the semantic relationships to the user. For example, the re-finder system may

display a multidimensional canvas showing representations of document clips and representations of concepts, in which the representations of document clips are positioned relative to locations of the representations of concepts according to the semantic relationships between the corresponding document clips and the corresponding concepts. The user may thus browse through document clips based on the semantic relationships between the document clips and the concepts, for example to find document clips that are semantically similar to and/or semantically dissimilar from respective ones of the concepts. In some aspects, the re-finder system may provide an interactive graphical user interface that may allow the user to interactively reposition the representations of concepts in the graphical user interface, and may update positions of the representations of document clips based on the changing locations of the representations of concepts in the graphical user interface. Thus, as the user is interactively repositioning a representation of a concept in the graphical user interface, representation of document clips that are relatively more semantically similar to the concept may be visually attracted to the representation of the concept in the graphical user interface and representations of document clips that are relatively less semantically similar to the concept may be visually repelled from the representation of the concept. The user may thus reposition the representations of concepts in the graphical user interface to organize the document clips according to document content that may conceptually be of interest to the user. These and other techniques described herein may enable the user to re-find documents that would have been more difficult or not possible to re-find using traditional re-finding techniques, such as using traditional keyword searches to locate documents having the content of interest to the user.

[0005] In aspects, a method for assisting users in re-finding documents is provided. The method includes generating embeddings for document clips related to respective documents among a plurality of documents. The method also includes receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents. The method additionally includes generating embeddings for respective concept clips in the first set of concept clips, and generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips. The method further includes determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document

clips and ii) the concept embedding, and generating a graphical user interface depicting the semantic relationships between the first concept and the document clips. The method additionally includes causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

[0006] In further aspects, a system is provided. The system for assisting users in re-finding documents is provided. The system comprises one or more computer readable storage media and

program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to perform operations. The operations include generating, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents. The operations also include receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents. The operations additionally include generating, using the machine learning model, embeddings for respective concept clips in the first set of concept clips, and generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips. The operations further include determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding, and generating a graphical user interface depicting the semantic relationships between the first concept and the document clips. The operations additionally include causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

[0007] In still further aspects, a computer storage medium is provided. The computer storage medium comprises computer executable instructions, which when executed by at least one processor, perform a method. The method includes generating embeddings for document clips related to respective documents among a plurality of documents. The method also includes receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents. The method further includes generating

embeddings for respective concept clips in the first set of concept clips, and generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips. The method further still includes determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding, and generating a graphical user interface depicting the semantic relationships between the first concept and the document clips. The method additionally includes causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

[0008] This Summary is provided to introduce a selection of concepts in a simplified form, which is further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the following description and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Non-limiting and non-exhaustive examples are described with reference to the following Figures.

[0010] Fig. 1 illustrates an overview of an example system in which re-finding techniques may be utilized, in accordance with aspects of the present disclosure.

[0011] Fig. 2 illustrates an example graphical user interface for providing user annotations for webpages, in accordance with aspects of the present disclosure.

[0012] Fig. 3 illustrates an example graphical user interface of a re-finder application, in accordance with aspects of the present disclosure.

[0013] Fig. 4 illustrates an example concept panel of a graphical user interface for creating concepts, in accordance with aspects of the present disclosure;

[0014] Fig. 5 illustrates an example concept panel of a graphical user interface for viewing and/or manipulating concepts, in accordance with aspects of the present disclosure;

[0015] Fig. 6 illustrates an example of a finder panel that may be provided as a detail panel of a graphical user interface, in accordance with aspects of the present disclosure;

[0016] Fig. 7 illustrates an example of a selected clip panel that may be provided as a detail panel of a graphical user interface, in accordance with aspects of the present disclosure;

[0017] Fig. 8 illustrates an example of a finder panel that may be provided as a detail panel of a graphical user interface, in accordance with aspects of the present disclosure;

[0018] Fig. 9 illustrates details of a method for assisting users in re-finding documents, in accordance with aspects of the present disclosure.

[0019] Fig. 10 is a block diagram illustrating physical components (e.g., hardware) of a computing device with which aspects of the disclosure may be practiced.

[0020] Figs. 11A-11B illustrate a mobile computing device with which aspects of the disclosure may be practiced.

DETAILED DESCRIPTION

[0021] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific aspects or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Aspects disclosed herein may be practiced as methods, systems, or devices. Accordingly, aspects disclosed herein may take the form of a hardware implementation, an entirely software implementation, or an implementation combining software and hardware aspects. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0022] In accordance with examples of the present disclosure, a re-finder system may be configured to assist users in re-finding documents having content of interest to the users. For example, the re-finder system may be configured to assist a user in re-finding webpages with particular content previously seen by the user. Although for purposes of ease of explanation, the re-finder system is generally described herein in the context of re-finding webpages previously seen by a user, the re-finder system may be configured to additionally or alternatively assist users in re-finding documents other than webpages previously seen by

the users. For example, the re-finder system may be configured to assist users in re-finding emails, chat message, documents stored in a file repository (e.g., document stored in a computer file system and/or on the cloud), etc. previously encountered by the users.

[0023] In aspects, the re-finder system may allow the user to define concepts that may include clips, such as parts of documents, images, text paragraphs, etc., that may be related to the content of interest to the user. The re-finder system may determine semantic relationships between the concepts defined by the user and clips of documents previously seen by the user, such as webpages in the browsing history of the user. In aspects, the re-finder system may generate (e.g., using a machine learning model, such as a transformer model though one of skill in the art will appreciate that other types of models may be used) semantic embeddings for the concepts defined by the user and for the clips of documents previously seen by the user, and may determine semantic relationships between the concepts and the document clips based on calculated distances between the semantic embeddings generated for the concepts and the semantic embeddings generated for the clips of documents previously seen by the user. The re-finder system may then display the relationships between the concepts and the document clips to the user. For example, the re-finder system may provide a graphical user interface displaying visual representations of the document clips and visual representations of the concepts, where the visual representations of the document clips are positioned relative to the visual representations of the concepts based on semantic relationships between the document clips and the concepts. The user may thus visually inspect documents based on the semantic similarities and/or dissimilarities between the document clips and the concepts to potentially re-find the document of interest to the user. In some aspects, the re-finder system may allow the user to interactively reposition the representations of concepts in the graphical user interface and may dynamically update locations of the representations of document clips based on the changing locations of the representations of the concepts in graphical user interface. Thus, as the user is interactively repositioning a representation of a concept in the graphical user interface, representations of document clips that are relatively more semantically similar to the concept may be visually attracted to the representation of the concept in the graphical user interface and representations of document clips that are relatively less semantically similar to the concept may be visually repelled from the representation of the concept in the graphical user interface. The user may thus reposition the

representations of concepts in the graphical user interface to organize the document clips according to document content that may conceptually be of interest to the user. These and other techniques described herein may enable the user to re-find documents that would have been more difficult or not possible to re-find using traditional re-finding techniques, such as using traditional keyword searches to locate documents having the content of interest to the user.

[0024] Fig. 1 illustrates an overview of an example system 100 in which re-finding techniques described herein may be utilized, in accordance with aspects of the present disclosure. The system 100 may include a user device 102 that may be configured to run or otherwise execute a re-finder application 104. Examples of the user device 102 include, but are not limited to, a laptop, a tablet, a smartphone, and the like. The re-finder application 104 may be a client application that may be configured to assist a user 110 of the user device 102 in re-finding electronic documents that may have been previously seen by the user 110. As an example, the re-finder application 104 may assist the user 110 in re-finding webpages previous seen by the user 110. In other examples, the re-finder application 104 may additionally or alternatively assist the user 110 in re-finding other types of documents, such as electronic file documents, emails, chats, etc. The re-finder application 104 be a client application that may be provided, for example, as an extension to a browsing application, such as a web browsing or a file browsing application. In some aspects, the re-finder application 104 may be a client application provided as an extension to another type of application in which searching for particular content may be needed or desired. For example, the re-finder application 104 may be provided as a part of, or an extension to, an email application, a chat application, a document repository application, etc. In other aspects, the re-finder application 104 may be a standalone application. In some examples, the re-finder application 104 may be a web application, where such application 104 may run or otherwise execute instructions within a web browser. In some examples, the application 104 may additionally or alternatively include a native client application residing on the user device 102.

[0025] The user device 102 may be communicatively coupled to an application server 106 via a communication network 108. The communication network 108 may be a wide area network (WAN) such as the Internet, a local area network (LAN), or any other suit able type

of network. The communication network 108 may be single network or may be made up of multiple different networks, in some examples. The system 100 may also include a database 112 that may be communicatively coupled to the application server 106 and/or to the user device 102 via the communication network 108. In other aspects, the database 112 may be coupled to the application server 106 and/or to the user device 102 in other suitable manners. For example, the database 112 may be directly connected to the application server 106, or may be included as part of the application server 106, in some examples. The database 112 may be a single database or may include multiple different databases.

[0026] In aspects, the re-finder application 104 may enable the user 110 to re-find documents (e.g., webpages) previously encountered by the user 110, for example, by accessing the document on the user device 102 or on another device, based on semantic relationships between document clips related to content of the documents previously encountered by the user 110 and concepts that may be defined or otherwise provided by the user 110. In this way, the re-finder application 104 may enable the user 110 to re-find the document without remembering the exact content of the document and without having to provide exact keywords for searching for the document. Although for purposes of ease of explanation, the re-finder application 104 is generally described herein in the context of re-finding webpages previously encountered by a user, the re-finder application 104 may be configured to additionally or alternatively assist users in re-finding documents other than webpages previously encountered by the users. For example, the re-finder application may be configured to assist users in re-finding emails, chat message, documents stored in a file repository (e.g., document stored in a computer file system and/or on the cloud), etc. previously encountered by the users.

[0027] In aspects, to enable the user 110 to subsequently re-find webpages encountered by the user 110, the re-finder application 104 may be configured to generate document clips 114 related to content of the webpages as the webpages are encountered by the user 110. In an aspect, the re-finder application 104 may include a document chunking module 116 configured to generate the document clips 114 for the webpages encountered by the user 110. The document clips 114 generated for a webpage may include, for example, document chunks containing text (e.g., text paragraphs) that the document chunking module 116 may parse out

from the webpage. In some aspects, the document chunking module 116 may discard some document chunks extracted from the webpage. For example, the document chunking module 116 may discard duplicate chunks containing same content extracted from the webpage. As another example, the document chunking module may discard document chunks that are too small, such as document chunks with a number of characters or words below a particular (e.g., predetermined) low threshold and/or document chunks that are too large, such as document chunks having a number of characters or words above a particular (e.g., predetermined) high threshold. In addition to or instead of text parsed out from a webpage, the document clips 114 generated for the webpage may include information other than text parsed out from the webpage. For example, the document clips 114 may include images extracted from the webpage. As another example, the document clips 114 may include information related to structure or feel of the webpage. For example, the document clips 114 may include metadata information such as information indicating a length of the webpage, information indicating a color (e.g., a background color) used on the webpage, information indicating types, colors, sizes, etc. of fonts used on the webpage, information indicating a number of sections or parts (e.g., tabs, etc.) on the webpage, etc.

[0028] In some aspects, the document clips 114 may include additional information that is not obtained from the webpage. For example, the re-finder application 104 may allow the user 110 to provide (e.g., enter into a graphical user interface 118 of the re-finder application 104) additional information, such as notes or other annotations, about the webpage, e.g., when the webpage is encountered by the user 110. The user 110 may thus add notes or other annotations, for example, that may help the user 110 to subsequently re-find the webpage. In examples, the re-finder application 104 may allow the user 110 to provide global annotations describing content and/or feel of the webpage as a whole and/or may allow the user 110 to provide local annotations describing content and/or feel of specific sections, paragraphs, images, etc. on the webpage. In aspects, the document chunking module 116 may include such annotations provided by the user 110 in the document clips 114 generated for the webpage. The document chunking module 116 may additionally or alternatively use the annotations provided by the user 110 to prioritize parts (e.g., chunks) of the webpage to be included in the document clips 114. For example, the user 110 may explicitly indicate importance of a particular part of the webpage when providing an annotation of the particular

part of the webpage and/or the document chunking module 116 may implicitly determine importance of a particular part of the webpage based on annotations provided by the user 110 for the particular part of the webpage. The document chunking module 116 may then generate the document clips 114 to include relatively more important parts of the webpage and exclude relatively less important parts of the webpage, such as parts that may not provide meaningful semantic information about the webpage.

[0029] The re-finder application 104 may provide the document clips 114 generated by the document chunking module 116 to the server 106 via the network 108. The server 106 may receive the document clips 114 and may provide the document clips 114 to a data service application 124 that may be running or otherwise executing on the server 106. The data service application 124 may include a document embedding module 126, a concept embedding module 128 and a concept search module 130. The document embedding module 126 may be configured to generate embeddings for the document clips 114 to capture semantics in the document clips 114. In an aspect, the document embedding module 126 may utilize an embedding model 132 to generate embeddings for the document clips 114. In aspects, the embedding model 132 may comprise a machine learning model, such as a transformer model, trained to generate embeddings based on content, where the embeddings capture semantics of the content. In some aspects, a multimodal transformer model may be utilized. The multimodal transformer model is trained based upon multiple modalities (e.g., webpages, file documents, emails chats, etc.) to generate embeddings for the various modalities in the same embedding space. In other aspects, the transformer model may be trained based upon a single modality, such as webpages for example. In some aspects, suitable machine learning models other than a transformer model may be utilized.

[0030] The data service application 124 may store the embeddings generated for the document clips 114 in the database 112 so that the embeddings may subsequently be used to determine semantic relationships between the corresponding document clips and concepts that may subsequently be provided by the user 110. In aspects, the data service application 124 may store the embeddings generated for the document clips 114 as part of document metadata 125 that may be included, for example, along with browsing history associated with the user 110 in the database 112. Subsequently, when the user 110 wishes to re-find a webpage

previously encountered by the user 110, the user 110 may use the re-finder application 104 to define or otherwise provide one or more concepts 134 for searching for the webpage that may, for example, be in the browsing history associated with the user 110. Such concepts may assist the user 110 to locate a particular webpage that may in a document repository 127 (e.g., in the database 112 or in a repository separate from the database 112), for example. In an aspect, the user 110 may provide a definition of a concept 134 via a concept panel 136 that may be displayed in the graphical user interface 118. In examples, a concept 134 may include one or more concept clips conceptually related to content of the webpage that the user 110 wishes to re-find. The one or more concept clips may include text that the user 110 may obtain from other documents, such as one or more text paragraphs that the user 110 may copy from one or more other webpages that may have been encountered by the user 110 and paste into the concept panel 136. The one or more concept clips may additionally or alternatively include one or more custom text paragraphs that may be entered by the user 110 into the concept panel 136 by the user 110, one or more images that may be provided via the concept panel 136 by the user 110, etc. In some aspects, the user 110 may obtain a concept definition from a concept repository, such as a library, that may store previously defined concepts such as standard concepts and/or saved concepts previously defined by the user 110 and/or users other than the user 110. In some aspects, the user 110 may customize such concepts, for example using the concept panel 136, to make the concepts suited for the search being currently performed by the user 110. As will be explained in more detail below, in various aspects, the user 110 may provide a definition of only a single concept for performing the search, or may provide respective definitions for multiple concepts for performing the search. As will also be explained in more detail below, the user 110 may change or modify concepts and/or may provide additional concepts to refine the search based on search results obtained based on concepts initially provided by the user 110.

[0031] The re-finder application 104 may provide the concepts 134 to the data service application 106 via the network 108. The data service application 106 may receive the concepts 134 and may provide the concepts 134 to the concept embedding module 128. The concepts embedding module 128 may generate embeddings for the concepts 134, where the embeddings may capture semantics of concept clips included in the concepts 134. In aspects, the concept embedding module 128 may generate an embedding for a concept 134 based on

a combination of embeddings that the concept embedding module 128 may generate for respective concept clips included in the concept 134. The embeddings for the concept clips may be generated by the concept embedding module 128 using a machine learning model, such as a transformer model, that may generate embeddings that capture semantics in the concept clips. In an aspect, the concept embedding module 128 may utilize the embedding model 132 to generate the embeddings for the concept clips included in the concept 134. In other aspects, a separate embedding model may be utilized for generating embeddings for the concept clips included in the concepts 134.

[0032] In aspects, the concept embedding module 128 may generate an embedding for a concept 134 as a linear combination of the embeddings generated for the concept clips included in the concept 134. In some aspects, the concept embedding module 128 may generate an embedding for a concept as a weighted linear combination of the embeddings generated for the concept clips included in the concept, where embeddings generated for some concept clips may be weighted higher than embeddings generated for other concept clips. The weights to be used for the embeddings may be provided along with the concept definition by the user 110, for example. In other aspects, the concept embedding module 128 may be configured to determine weights to be used for the embeddings in other suitable manners. In some examples, the concept embedding module 128 may generate an embedding for a concept 134 as an unweighted or equally weighted linear combination of the embeddings generated for the concept clips included in the concept 134 and/or may generate an embedding for a concept 134 based on a non-linear combination of the embeddings generated for the concept clips included in the concept 134.

[0033] The concept embeddings generated for the concepts 134 by the concept embedding module 128 may be provided to the concept search module 130. The concept search module 130 may be configured to obtain (e.g., retrieve from the database 112) embeddings generated by the document embedding module 126 for the document clips 114 related to webpages previously encountered by the user 110, and determine semantic relationships between the concepts 134 and the document clips 114 based on i) respective embeddings generated by the document embedding module 126 for the document clips 114 and ii) the concept embedding generated by the concept embedding module 128 for the concepts 134. For example, the

concept search module 130 may be configured to determine semantic relationships between a concept 134 and the document clips 114 based on calculated distances between i) respective embeddings generated by the document embedding module 126 for the document clips 114 and ii) a concept embedding generated by the concept embedding module 128 for the concept 134.

[0034] The concept search module 130 may provide indications of the determined semantic relationships between the concepts 134 and the document clips 114 to the re-finder application 104. In some aspects, the concept search module 130 may generate a concept canvas for displaying semantic relationships between the concepts 134 and the document clips 114, and may provide the concept canvas display to the re-finder application 104 for display to the user 110. The concept canvas display may comprise a multi-dimensional (e.g., two-dimensional) canvas or map in which representations of the concepts 134 and representations of the document clips 114 may be laid out to display semantic relationships between the concepts 134 and the document clips 114. As will be explained in more detail below, the concept search module 130 may generate a graph having nodes corresponding the document clips 114 and the concepts 134 and edges that represent semantic relationships between the document clips 114 and the concepts 134. As will also be explained in more detail below, the concept search module 130 may then determine a layout for displaying the semantic relationships between the document clips 114 and the concepts 134 using a force-directed graph drawing algorithm based on the graph, where nodes with relatively stronger connections to each other in the graph are attracted to each other with relatively greater forces in the layout of the graph. In other aspects, the layout for displaying the semantic relationships between the document clips 114 and the concepts 134 may be determined in other suitable manners.

[0035] The re-finder application 104 may render the concept canvas to the user 110 in a canvas panel 154 of the graphical user interface 118. In aspects, the concept panel 154 may thus display representations of the document clips 114 and representations of the concepts 134, where the representations of the document clips 114 may be positioned in relation to representations of the concepts 134 based on the semantic relationships between the document clips 114 and the concepts 134. Thus, for example, a representation of a document clips 114

that is semantically more similar to a concept 134 may be placed in the canvas panel 154 relatively closer to a representation of the concept 134 in the canvas panel 154 as compared to a representation of a document clip 114 that is semantically less similar to the concept 134. The user 110 may then browse through the document clips 114 according to the semantic similarity of the document clips 114 to the concepts 134 to locate one or more document clips, e.g., a subset of a documents clip 114, signifying that the one or more document clips may be associated with a webpage that the user 110 is attempting to re-find. In some aspects, the graphical user interface 118 may allow the user 110 to click on, hover over, or otherwise engage with representations of document clips 114 in the canvas panel 154 to view content of the document clips 114 to browse through the document clips 114. In other aspects, the graphical user interface 118 may allow the user 110 to browse through the document clips 114 in other suitable manners. In aspects, the graphical user interface 118 may display webpages (e.g., retrieved from the document repository 127) associated with the document clips 114 selected by the user 110, to allow the user 110 to browse through the webpages associated with the document clips 114.

[0036] In some aspects, the graphical user interface 118 may be operable to enable the user 110 to interactively reposition representations of the concepts 134 in the canvas panel 154. For example, the graphical user interface 118 may be operable to enable the user 110 to drag a representation of a concept 134 to a new location in the canvas panel 154. As the user 110 is repositioning the representation of the concept 134 in the canvas panel 154, the re-finder application 114 may dynamically update locations of the representations of the document clips 114 in the canvas panel 154 based on the changing location of the representation of the concept 134. Thus, as the user 110 is repositioning a representation of a concept 134 in the concept panel 154, the representation of document clips 114 that are relatively more semantically similar to the concept 134 may be visually attracted to the representation of the concept 134 in the concept panel 154 and representations of document clips that are relatively less semantically similar to the concept 154 may be visually repelled from the representation of the concept 134 in the concept panel 154. The user may thus reposition the representations of concepts 134 in the graphical user interface 118 to organize the document clips 114 according to content that may be conceptually of interest to the user. In some aspects, the re-finder application 104 may allow the user 110 to modify the concepts 134 based on the search

results displayed in the concept canvas 154. For example, the user 110 may modify content of, delete and/or adjust importance (e.g., weights) of one or more concept clips included in a concept 134 and/or may provide one or more additional concept clips to be included in the concept 134. As another example, the user 110 may entirely delete one or more concept 134 and/or may add one or more additional concepts 134. The re-finder application 104 may update the display in the canvas panel 154 based on the modified concept definitions provided by the user 110. These and other techniques described herein may enable the user 110 to re-find documents that would have been more difficult or not possible to re-find using traditional re-finding techniques, such as using a traditional keyword searches to locate documents having the content of interest to the user.

[0037] Referring still to Fig. 1, it should be appreciated that while the document chunking module 116 is illustrated in Fig. 1 as being a part of a re-finder application executing or otherwise running on the user device 102, the document chunking module 116 may be at least partially executed on the server 106 in some aspects. Similarly, it should be appreciated that while the document embedding module 126, the concept embedding module 128 and the concept search module 130 are illustrated in Fig. 1 as being parts of a data service application executing or otherwise running on the server 106, the document embedding module 126, the concept embedding module 128 and/or the concept search module 130 may be at least partially executed at the re-finder application 104 of the user device 102 in some aspects. For example, the document embedding module 126 may at least partially operate to generate embeddings locally at the re-finder application 104 executing or otherwise running on the user device 102. Moreover, the concept embedding module 128 and/or the concept search module 130 may at least partially reside at the re-finder application executing or otherwise running on the user device 102.

[0038] Fig. 2 illustrates an example of a graphical user interface 200 configured to enable a user to provide document (e.g., webpage) annotations, in accordance with aspects of the present disclosure. The graphical user interface 200 may be utilized with the system 100 of Fig. 1, and the graphical user interface 200 is described for exemplary purposes with reference to the system 100 of Fig. 1. In other aspects, the graphical user interface 200 may be utilized with systems different from the system 100 of Fig. 1.

[0039] The graphical user interface 200 may be provided as an extension to an internet search application, such as an internet browser, for example. In an aspect, the graphical user interface 200 may be displayed on the user device 102 of Fig. 1 as the user 110 visits webpage using a browser executing or otherwise running on the user device 102. The graphical user interface 200 may include a bar (e.g., a sidebar) 202 for providing notes or other annotations for webpages encountered by the user 110. The bar 202 may be displayed to the user 110 when the user 110 clicks on or otherwise engages with an icon 204 that may be provided, for example, as extension to the browser being utilized by the user 102. The bar 202 may include an area (e.g., a box) 206 for entering custom information, such as notes or other annotations that the user 110 may wish to add about a webpage to facilitate subsequent re-finding of the webpage. In an aspect, such notes may be included in document clips 114 associated with the webpage. For example, the document chunking module 116 may include the note or other annotation provided by the user 110 as one or more document clips 114 generated for the webpage, along with one or more document clips 114 generated based on the content of the webpage. In this way, the custom information entered by the user 110 may subsequently be used in a semantic concept search assist the user 110 in re-finding the webpage.

[0040] Fig. 3 illustrates an example graphical user interface 300 of a re-finder application, in accordance with aspects of the present disclosure. In an aspect, the graphical user interface 300 corresponds to the graphical user interface 118 of the re-finder application 104 of Fig. 1. In another aspect, the graphical user interface 300 may be provided by a re-finder application different from the re-finder application 104 of Fig. 1.

[0041] The graphical user interface 300 may include a concept panel 302, a canvas panel 304 and a detail panel 306. In aspects, the concept panel 302 may allow a user to create and/or manipulate one or more concepts 310 for searching through browsing history of the user. Referring briefly to Fig. 4, the concept panel 302 may include a concept title field 412, a drop box 414 and a custom text box 416 that may be utilized for creation of a concept 310. The user may enter a title for the concept 310 into the concept field 410, and may define one or more concept clips for the concept 310 using the drop box 414 and/or the custom text box 416. For example, the user may use the drag and drop box 414 to copy text or image from another document, such as another webpage, the user may wish to be included as a concept

clip in the concept 310. The user may additionally or alternatively enter custom text, such as a written paragraph to be included as a concept clip in the concept 310, into the custom text box 414 for the concept 310. In some aspects, the concept panel 302 may additionally allow the user to specify weights to be used for each of the one or more concept clips. The weights may signify relative importance of the respective concept clips to the concept 310. In aspects, weights may be utilized when generating a concept embedding based on embeddings generated for the corresponding concept clips.

[0042] Referring now briefly to Fig. 5, the concepts 310 created by the user may be displayed as expandable menus 510 in the concept panel 302. The user may click on a plus sign associated with a concept 310 to expand the view of the concept 310 in the concept panel 302. The user may then manipulate one or more concept clips in the expanded view of the concept 310. For example, the user may modify content of and/or delete one or more concept clips from the concept 310. As another example, the user may adjust weights of one or more concept clips included in the concept 310. As yet another example, the user may provide one or more additional concept clips for the concept 310. In some aspects, the user may additionally or alternatively entirely delete one or more concepts 310 from the search and/or add one or more additional concepts 310 to the search. In aspects, as will be explained in more detail below, such manipulations of the concepts 310 may be performed by the user to refine an initial search performed based on initially defined one or more concepts 310 by the user.

[0043] Referring back to Fig. 3, the canvas panel 304 may display representations of the concepts 310 provided for a search by the user and representations of clips 312 that may be semantically similar to respective ones of the concepts 310 provided by the user. The representations of clips 312 may include representation of document clips (e.g., document clips 114 of Fig. 1) generated for webpages in the browsing history associated with the user. The clips 312 may additionally include concept clips included in the concepts 310. In the example illustrated in Fig. 3, concepts 310 are represented by larger squares in the canvas panel 304 and clip 312 are presented by the smaller squares in the canvas panel 304. In some aspects, representations of concept clips included in a concept 310 may be visually displayed

as related to the representation of the concept 310. For example, representations of concept clips included in a concept 310 may be of same color as the representation of the concept 310.

[0044] In an aspect, the representations of the clips 312 may be positioned in relation to each other in the canvas panel 304 based on degrees of semantic similarity between the clips 312 and may further be arranged based on degrees of semantic similarities between the clips 312 and the concepts 310 in the canvas panel 304. For example, locations of the representations of the clips 312 in the canvas panel 304 may be determined based on distances between embeddings generated (e.g., by the document embedding module 126 and/or concept embedding module 128 of Fig. 1) for the respective clips 312 and further based on distances between the embedding generated for the clips 312 and embeddings generated (e.g., by the concept embedding module 128 of Fig. 1) for the concepts 310. The user may thus view clips 312 that may be semantically similar each other and further semantically similar to multiple concepts 310 and/or semantically more similar to a particular concept 310 than other ones of the concepts 310. The user may thus attempt to locate a webpage based on semantic similarities and/or dissimilarities of clips 312 associated with the webpage to respective ones of the concepts 310. In the example illustrated in Fig. 3, the user has provided four concepts 310 for the search, including a “foods and drinks” concept 310-1, a “nutrition” concept 310-2, a “cooking” concept 310-3 and a “vegetarian meals” concept 310-4. If the user is attempting to re-find a webpage about both nutrition and feed and drink, the user may wish to view clips 312 that are positioned in an area in the canvas panel 304 that may be at least substantially equidistant between the position of the concept “nutrition” 310-1 and the concept “food and drink” 310-2 in the canvas panel 304. On the other hand, if the user is attempting to find a webpage that is about food and drink but not about nutrition, then the user may wish to view clips 312 that are positioned in an area in the canvas panel 304 that is closer to the concept “food and drink” 310-2 and farther away from the concept “nutrition” in the canvas panel 304.

[0045] In some aspects, the user may wish to refine a search by manipulating (e.g., changing or deleting) one or more concepts 310 initially provided by the user and/or by providing one or more additional concepts 310. As an example, the concept “nutrition” 310-2 may be provided by the user after viewing initial search results in the canvas panel 304 in order to

“pull way” or separate clips 312 that are semantically more closely related to nutrition from the clips 312 that are more semantically related to the concept “food and drink” but not “nutrition.” In response to the user modifying, deleting and/or adding a concept 310, the re-finder application 104 may update the canvas panel 304 based on semantic relationships determined based on the clips 312 and the new definition of the concepts 310. For example, if a representation of a first concept (e.g., the “food and drink” concept 310-1) is placed at a first location in the canvas panel 304 and subsequently a representation of a second concept (e.g., the “nutrition” concept 310-2) is placed at a second location in the canvas panel 304, updated locations of the representations of the clips 312 in the canvas panel 304 may be determined based on i) the first location of the representation of the first concept (e.g., the “food and drink” concept 310-1) in the in the canvas panel 304, ii) calculated distances between the embeddings generated for the corresponding clips 312 and the first concept embedding, iii) the second location of the representation of the second concept (e.g., the “nutrition” concept 310-2) in the in the canvas panel 304 and iv) calculated distances between the embeddings 312 generated for the corresponding clips and the second concept embedding. The re-finder application 104 may then move the representations of clips 312 to the updated locations. The user may thus more easily locate a webpage if the webpage that the user is attempting to re-find is related to the first concept (e.g., food and drink) but not the second concept (e.g., nutrition).

[0046] In an aspect, the positions of the representations of the clips 312 relative to each other in the canvas panel 304 may be determined (e.g., by the concept search module 130 of Fig. 1) based on an undirected graph of the clips 312. In some aspects, types of graphs other than an undirected graph may be utilized. For example, in an aspect, a directed graph may be utilized. In generating a graph of the clips 312, the concept search module 130 may place the clips 312 as nodes in the graph and may add edges based on distances between embeddings generated for the clips 312. In an aspect, the concept search module 130 may add edges between nodes corresponding to clips 312 having similarity (e.g., cosine similarity) greater than a particular (e.g., predetermined) threshold. For example, the concept search module 130 may add edges between nodes corresponding to clips 312 having similarity (e.g., cosine similarity) greater than zero. In another aspect, a suitable threshold greater than zero may be utilized. In some aspects, the concept search module 130 may perform sampling of edges in

the graph in order to reduce the number of edges and, accordingly, to reduce the complexity of the graph. For example, for each node in the graph, the concept search module 130 may reduce the number of edges that connect to the node such that the total number of edges connecting to the node is $k\%$ of all edges connecting to all of the nodes in the graph. In this way, the concept search module 130 may proportionally sample $k\%$ of all edges in the graph. Thus, in an aspect, if the number of edges in the graph $O(N^2)$, then the number of sampled edges is $k/100 * O(N^2)$. In an example, the value of k is 1. In this example, the concept search module 130 may proportionally sample 1% of all edges in the graph. In at least some aspects, this sampling may provide a balanced trade-off between computational time and power needed to determine the edges and accurate representation of the relationships between the nodes, so that locations of the representations of the clips 312 may be updated with responsive, interactive rates when the user 110 is repositioning a concept 310 on the canvas panel 304, for example. In other aspects, other suitable values of k are utilized and, according, other suitable percentages of all edges are sampled.

[0047] In an aspect, the concept search module 130 may compute a layout for displaying the representations of the clips 312 in the canvas panel 304 using a force-directed graph drawing algorithm based on the graph, where nodes with relatively stronger connections to each other in the graph are attracted to each other with relatively greater forces in the layout of the graph. In this way, representations of clips 312 that are relatively more similar to each other are positioned closer to each other in the canvas panel 304, and representations of clips 312 with sufficiently large degrees of similarity may form clusters in the canvas panel 304. To provide a visualization of similarities between clips 312 and concepts 310 in the canvas panel 304, the concept search module 130 may further add concepts 310 as nodes to the underlying graph used for computing the layout of the display in the canvas panel 304. Upon adding a concept 310 as a node to the graph, the concept search module 130 may connect the node corresponding to the concept 310 to other nodes in the graph based on semantic similarity between the concept 310 and the clips 312 corresponding to the other nodes in the graph. In an aspect, the concept search module 130 connects the node corresponding to the concept 310 to m other nodes in the graph with relatively greater semantic similarities between the concept 310 and the clips 312 corresponding to the other nodes in the graph. In an example, the value of m is 25. In this example, the concept search module 130 connects the node corresponding

to the concept 310 to 25 other nodes in the graph with relatively greater semantic similarities between the concept 310 and the clips 312 corresponding to the other nodes in the graph. Connecting each node corresponding to each concept 312 to 25 other nodes in the graph with relatively greater semantic similarities between the concept 310 and the clips 312 corresponding to the other nodes in the graph may provide a balanced trade-off between computing time and usefulness of the layout of the graph.

[0048] In aspects, the concept search module 130 may generally treat locations of the representations of concepts 310 as fixed locations on the canvas panel 304 and may use the fixed locations of the representations of concepts 310 as constraints to the force-directed algorithm when determining the layout of the graph in the canvas panel 304. However, as discussed above, the re-finder application 104 may allow the user to interactively move (e.g., drag) the representations of the concepts 310 across the canvas panel 304. As the user interactively moves a representation of a concept 310 to a new location in the canvas panel 304, the re-finder application 104 may update locations of the representations of clips 312 that are connected directly and indirectly to the node corresponding to the concept 310 based on the changing location of the representation of the concept 310. In an aspect, the connections between the node corresponding to the concept 310 may be dynamically updated such that m other nodes with relatively greater semantic similarities to the concept 310 move closer to the new location of the representation of the concept 310. Thus, in an aspect, as the user moves a representation of a concept 310 across the canvas panel 304, representations of clips 312 that are relatively more semantically similar to the concept 310 appear to be attracted to the representation of the concept 310 in the canvas panel 304.

[0049] In an aspect, the re-finder application 104 may enable the user to view content of the clips 312 in the canvas panel 304, for example by hovering over or otherwise engaging with a representation of the clips 312 in the canvas panel 304. By browsing through the clips 312 based on the locations of the clips 312 relative to the concepts 310, the user may recognize that a particular clip 312 belongs to a webpage that the user is attempting to re-find. In some aspects, the user may then click on or otherwise engage with the clip 312 to be directed to the webpage.

[0050] In some aspects, the re-finder application 104 may display details of one or more clips 312 that may be of interest to the user in the detail panel 306. For example, the user may select an area of interest to the user using a finder tool 344, and the detail panel 306 may display a list of one or more clips 312 that are located in the canvas panel 304 in the area selected by the user. In some aspects, the user may click on or otherwise engage with a representation of a particular clip 312 in the canvas panel 304 to cause a view displaying content of the particular clip 312 to be displayed in the detail panel 306. In some aspects, the detail panel 306 may allow the user to perform keyword query search of the content of clips 312 associated with webpages in the browsing history of the user. In this view, the detail panel 306 may display all clips 312 that include the query provided by the user.

[0051] In some aspects, when content of a clip 312 is displayed in the detail panel 304, one or more keywords in the clip 312 may be highlighted to the user. In an aspect, the one or more keywords to be highlighted in a clip 312 may be determined using a machine learning model, such as a transformer based machine learning model, that may identify keywords having corresponding embeddings that are sufficiently close to the embedding of the clip 312. For example, the model may identify one or more keywords in a clip 312 that have embeddings with highest cosine similarity with the embedding of the clip 312. Highlighting identified keywords in a clip 312 in the detail panel 306 may assist the users to more quickly assess importance of the clip 312 and relevancy of content of the clip 312 to the one or concepts 310 provided by the user.

[0052] Fig. 6 illustrates an example of a finder panel 600 that may be provided as the detail panel 306, in accordance with aspects of the present disclosure. The re-finder application 104 may display the finder panel in response to detecting that the user has positioned the finder tool 344 over a particular area in the canvas panel 304. The finder panel 600 may display content of one or more clips 312 that are located in the particular area in the canvas panel 304 selected by the user. In an aspect, content of clips 312 in the finder panel 600 may be displayed in association with the webpages of which the clips 312 may be a part. For example, content of each clip 312 in the finder panel 600 may be displayed on a respective page card 602 corresponding to the webpage of which the clip 312 is a part. In some aspects, respective page cards 602 in the finder panel 600 may include additional information about the

corresponding webpages, such as a title of the webpage, a date and time of the user's last visit to the webpage, etc. In an aspect, if the user wishes to visit a webpage associated with a clip 312, the user may click on a portion of the page card 602 (e.g., displaying the title of the webpage) corresponding to the webpage, and the re-finder application 104 may direct the user to the webpage.

[0053] Fig. 7 illustrates an example of a selected clip panel 700 that may be provided as the detail panel 306, in accordance with aspects of the present disclosure. The re-finder application 104 may display the selected clip panel 700 in response to detecting that the user has clicked on or otherwise engaged with a particular clip 312 in the canvas panel 304. In an aspect, the selected clip panel 700 may display content of the particular clip 312 selected by the user. In some aspects, the selected clip panel 700 may additionally display content of clips 312 that are closely related to the particular clip 312 selected by the user. For example, the selected clip panel 700 may display a predetermined number of clips 312 that are the closest members of the selected clip 312. In an aspect, content of each clip 312 displayed in the selected clip panel 700 may be displayed in association with the webpage associated with the clip 312. For example, as with the finder user interface 600 in Fig. 6, content of each clip 312 in the selected clip panel 700 may be displayed on a respective page card 702. In some aspects, respective page cards 702 in the selected clip panel 700 may include additional information about the corresponding webpages, such as a title of the webpage, a date and time of the user's last visit to the webpage, etc. In an aspect, if the user wishes to visit a webpage associated with a clip 312, the user may click on a portion of the page card 702 (e.g., displaying the title of the webpage) corresponding to the webpage, and the re-finder application 104 may direct the user to the webpage.

[0054] Fig. 8 illustrates an example of a keyword search panel 800 that may be provided as the detail panel 306, in accordance with aspects of the present disclosure. In an aspect, the re-finder application 104 may display the keyword search panel 800 in response to detecting that the user has clicked or otherwise engaged with a search icon (not shown) that may be displayed in the detail panel 306. The keyword search panel 800 may include a search box 804 that may allow the user to enter keyword search queries for searching through the browsing history of the user. In response to the user entering a search query in the search box

804, the keyword search panel 800 may display clips 312, associated with webpages in the browsing history of the user, that include text similar to and/or the exact wording of the query. In an aspect, clips 312 that are associated with a particular webpage may be displayed in the keyword search panel 800 on a page card 806 corresponding to the particular webpage. The page card 806 may include additional information about the webpage, such as a title of the webpage, a date and time of the user's last visit to the webpage, etc. In an aspect, if the user wishes to visit the webpage, the user may click on a portion of the page card 806 (e.g., displaying the title of the webpage), and the re-finder application 104 may direct the user to the corresponding webpage.

[0055] Fig. 9 illustrates details of a method 900 for assisting users in re-finding documents, in accordance with aspects of the present disclosure. A general order for the steps of the method 700 is shown in Fig. 9. The method 900 can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer readable medium. Further, the method 900 can be performed by gates or circuits associated with a processor, Application Specific Integrated Circuit (ASIC), a field programmable gate array (FPGA), a system on chip (SOC), or other hardware device. Hereinafter, the method 900 shall be explained with reference to the systems, components, modules, software, data structures, user interfaces, etc. described in conjunction with Figs. 1-8.

[0056] At block 902, embeddings are generated for document clips related to respective documents among a plurality of documents. In an aspect, the document clips related a particular document among the plurality of documents include one or more of i) one or more text paragraphs parsed out from the particular document and ii) one or more images from the particular document. In some aspects, the document clips related to the particular document additionally or alternatively include information related to document metadata information, such as information about structure and/or feel (e.g., length of document, number and/or lengths of respective sections (e.g., paragraphs) of the particular document, fonts (e.g., font type, size, color, etc.) used in the particular document, document background color, etc.. In some aspects, the document clips related to a particular document include at least one user-defined clip that includes information not extracted from the document. In an aspect, the

embeddings for the document clips are generated using a machine learning model, such as a transformer model, configured to capture semantics.

[0057] At block 904, a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents. In an aspect, the first set of clips defining the first concept includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user. While specific types of content are described herein, one of skill in the art will appreciate the set of clips may include other types of content, such as, for example, video or audio clips.

[0058] At block 906, embeddings for respective concept clips in the first set of concept clips are generated. In an aspect, the embeddings for the concept clips are generated using a machine learning model, such as a transformer model, configured to capture meaning of content. While aspects are described as using a transformer model, one of skill in the art will appreciate that other types of machine learning models may be employed with the disclosed aspects. In an example, the embeddings for the concept clips at block 906 are generated using the same machine learning model as used for generating the embeddings for the document clips at block 902. In another example, a different machine learning model is utilized.

[0059] At block 908, a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips. In an example, the first concept embedding is generated as a weighted linear combination of the embeddings for the concept clips generated at block 906. In other example, the first concept embedding based on another suitable combination of the embeddings for the concept clips generated at block 906

[0060] At block 910, semantic relationships between the first concept and the document clips are determined based on i) the respective embeddings generated for the document clips and ii) the concept embedding. For example, the semantic relationships are determined based on respective distances (e.g., cosine similarity distances) between i) the respective embeddings generated for the document clips and ii) the concept embedding.

[0061] At block 912, a graphical user interface depicting the semantic relationships between the first concept and the document clips is generated. In aspects, generating the graphical user interface at block 912 includes placing representations of the document clips in a multi-dimensional canvas, wherein the representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips. In aspects, generating the graphical user interface further includes placing a representation of the first concept at a first location in the multidimensional canvas, determining locations of the representations of the document clips relative to the first location of the representation of the first concept based on calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding, and positioning the representations of the document clips in relation to the representation of the first concept in the multidimensional canvas in accordance to the determined distances. In other aspects, the graphical user interface depicting the semantic relationships between the first concept and the document clips is generated in other suitable manners. At block 914, the graphical user interface is rendered at a client device. The graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

[0062] Figs. 10-11 and the associated descriptions provide a discussion of a variety of operating environments in which aspects of the disclosure may be practiced. However, the devices and systems illustrated and discussed with respect to Figs. 10-11 are for purposes of example and illustration and are not limiting of a vast number of computing device configurations that may be utilized for practicing aspects of the disclosure, described herein.

[0063] Fig. 10 is a block diagram illustrating physical components (e.g., hardware) of a computing device 1000 with which aspects of the disclosure may be practiced. The computing device components described below may be suitable for the computing devices described above. In a basic configuration, the computing device 1000 may include at least one processing unit 1002 and a system memory 1004. Depending on the configuration and type of computing device, the system memory 1004 may comprise, but is not limited to, volatile storage (e.g., random access memory), non-volatile storage (e.g., read-only memory), flash memory, or any combination of such memories.

[0001] The system memory 1004 may include an operating system 1005 and one or more program modules 1006 suitable for running software application 1020, such as one or more components supported by the systems described herein. As examples, system memory 1004 may store a document chunking module 1016 (e.g., corresponding to the document chunking module 126 of Fig. 1), a document embedding module 1026 (e.g., corresponding to the document embedding module 126 of Fig. 1), a concept embedding module 1028 (e.g., corresponding to the concept embedding module 128 of Fig. 1) and/or a concept search module 1030 (e.g., corresponding to the concept search module 130 of Fig. 1). The operating system 1005, for example, may be suitable for controlling the operation of the computing device 1000.

[0002] Furthermore, aspects of the disclosure may be practiced in conjunction with a graphics library, other operating systems, or any other application program and is not limited to any particular application or system. This basic configuration is illustrated in Fig. 10 by those components within a dashed line 1008. The computing device 1000 may have additional features or functionality. For example, the computing device 1000 may also include additional data storage devices (removable and/or non-removable) such as, for example, magnetic disks, optical disks, or tape. Such additional storage is illustrated in Fig. 10 by a removable storage device 1009 and a non-removable storage device 1010.

[0003] As stated above, a number of program modules and data files may be stored in the system memory 1004. While executing on the at least one processing unit 1002, the program modules 1006 (e.g., application 1020) may perform processes including, but not limited to, the aspects, as described herein. Other program modules that may be used in accordance with aspects of the present disclosure may include electronic mail and contacts applications, word processing applications, spreadsheet applications, database applications, slide presentation applications, drawing or computer-aided application programs, etc.

[0004] Furthermore, aspects of the disclosure may be practiced in an electrical circuit comprising discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. For example, aspects of the disclosure may be practiced via a system-on-a-chip (SOC) where each or many of the components illustrated in Fig. 10 may be

integrated onto a single integrated circuit. Such an SOC device may include one or more processing units, graphics units, communications units, system virtualization units and various application functionality all of which are integrated (or “burned”) onto the chip substrate as a single integrated circuit. When operating via an SOC, the functionality, described herein, with respect to the capability of client to switch protocols may be operated via application-specific logic integrated with other components of the computing device 1000 on the single integrated circuit (chip). Aspects of the disclosure may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including but not limited to mechanical, optical, fluidic, and quantum technologies. In addition, aspects of the disclosure may be practiced within a general purpose computer or in any other circuits or systems.

[0005] The computing device 1000 may also have one or more input device(s) 1012 such as a keyboard, a mouse, a pen, a sound or voice input device, a touch or swipe input device, etc. The output device(s) 1014 such as a display, speakers, a printer, etc. may also be included. The aforementioned devices are examples and others may be used. The computing device 1000 may include one or more communication connections 1017 allowing communications with other computing devices 1050. Examples of suitable communication connections 1017 include, but are not limited to, radio frequency (RF) transmitter, receiver, and/or transceiver circuitry; universal serial bus (USB), parallel, and/or serial ports.

[0006] The term computer readable media as used herein may include computer storage media. Computer storage media may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, or program modules. The system memory 1004, the removable storage device 809, and the non-removable storage device 1010 are all computer storage media examples (e.g., memory storage). Computer storage media may include RAM, ROM, electrically erasable read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other article of manufacture which can be used to store information and which can be accessed by the computing device 1000. Any such computer storage media may be part of the

computing device 1000. Computer storage media does not include a carrier wave or other propagated or modulated data signal.

[0064] Communication media may be embodied by computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” may describe a signal that has one or more characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, radio frequency (RF), infrared, and other wireless media.

[0007] Figs. 11A-11B illustrate a mobile computing device 1100, for example, a mobile telephone, a smart phone, wearable computer (such as a smart watch), a tablet computer, a laptop computer, and the like, with which aspects of the disclosure may be practiced. In some aspects, the client (*e.g.*, computing system 104A-E) may be a mobile computing device. With reference to Fig. 11A, one aspect of a mobile computing device 1100 for implementing the aspects is illustrated. In a basic configuration, the mobile computing device 1100 is a handheld computer having both input elements and output elements. The mobile computing device 1100 typically includes a display 1105 and one or more input buttons 1110 that allow the user to enter information into the mobile computing device 1100. The display 1105 of the mobile computing device 1100 may also function as an input device (*e.g.*, a touch screen display). If included, an optional side input element 1115 allows further user input. The side input element 1115 may be a rotary switch, a button, or any other type of manual input element. In alternative aspects, mobile computing device 1100 may incorporate more or less input elements. For example, the display 1105 may not be a touch screen in some aspects. In yet another alternative aspect, the mobile computing device 1100 is a portable phone system, such as a cellular phone. The mobile computing device 1100 may also include an optional keypad 1135. Optional keypad 1135 may be a physical keypad or a “soft” keypad generated on the touch screen display. In various aspects, the output elements include the display 1105 for showing a graphical user interface (GUI), a visual indicator 1120 (*e.g.*, a light emitting diode), and/or an audio transducer 1125 (*e.g.*, a speaker). In some aspects, the mobile

computing device 1100 incorporates a vibration transducer for providing the user with tactile feedback. In yet another aspect, the mobile computing device 1100 incorporates input and/or output ports, such as an audio input (e.g., a microphone jack), an audio output (e.g., a headphone jack), and a video output (e.g., a HDMI port) for sending signals to or receiving signals from an external source.

[0008] Fig. 11B is a block diagram illustrating the architecture of one aspect of computing device, a server, or a mobile computing device. That is, the computing device 1100 can incorporate a system (e.g., an architecture) 1102 to implement some aspects. The system 1102 can implemented as a "smart phone" capable of running one or more applications (e.g., browser, e-mail, calendaring, contact managers, messaging clients, games, and media clients/players). In some aspects, the system 1102 is integrated as a computing device, such as an integrated personal digital assistant (PDA) and wireless phone.

[0009] One or more application programs 1166 may be loaded into the memory 1162 and run on or in association with the operating system 1164. Examples of the application programs include phone dialer programs, e-mail programs, personal information management (PIM) programs, word processing programs, spreadsheet programs, Internet browser programs, messaging programs, and so forth. The system 1102 also includes a non-volatile storage area 1168 within the memory 1162. The non-volatile storage area 1168 may be used to store persistent information that should not be lost if the system 1102 is powered down. The application programs 1166 may use and store information in the non-volatile storage area 1168, such as e-mail or other messages used by an e-mail application, and the like. A synchronization application (not shown) also resides on the system 1102 and is programmed to interact with a corresponding synchronization application resident on a host computer to keep the information stored in the non-volatile storage area 1168 synchronized with corresponding information stored at the host computer. As should be appreciated, other applications may be loaded into the memory 1162 and run on the mobile computing device 1100 described herein (e.g., document chunking module, document embedding module, concept embedding module, concept search module, , etc.).

[0010] The system 1102 has a power supply 1170, which may be implemented as one or more batteries. The power supply 1170 might further include an external power source, such as an AC adapter or a powered docking cradle that supplements or recharges the batteries.

[0011] The system 1102 may also include a radio interface layer 1172 that performs the function of transmitting and receiving radio frequency communications. The radio interface layer 1172 facilitates wireless connectivity between the system 1102 and the "outside world," via a communications carrier or service provider. Transmissions to and from the radio interface layer 1172 are conducted under control of the operating system 1164. In other words, communications received by the radio interface layer 1172 may be disseminated to the application programs 1166 via the operating system 1164, and vice versa.

[0012] The visual indicator 1120 may be used to provide visual notifications, and/or an audio interface 1174 may be used for producing audible notifications via the audio transducer 1125. In the illustrated configuration, the visual indicator 1120 is a light emitting diode (LED) and the audio transducer 1125 is a speaker. These devices may be directly coupled to the power supply 1170 so that when activated, they remain on for a duration dictated by the notification mechanism even though the processor 1160 and other components might shut down for conserving battery power. The LED may be programmed to remain on indefinitely until the user takes action to indicate the powered-on status of the device. The audio interface 1174 is used to provide audible signals to and receive audible signals from the user. For example, in addition to being coupled to the audio transducer 1125, the audio interface 1174 may also be coupled to a microphone to receive audible input, such as to facilitate a telephone conversation. In accordance with aspects of the present disclosure, the microphone may also serve as an audio sensor to facilitate control of notifications, as will be described below. The system 1102 may further include a video interface 1176 that enables an operation of an on-board camera 1130 to record still images, video stream, and the like.

[0013] A mobile computing device 1100 implementing the system 1102 may have additional features or functionality. For example, the mobile computing device 1100 may also include additional data storage devices (removable and/or non-removable) such as, magnetic disks, optical disks, or tape. Such additional storage is illustrated in Fig. 11B by the non-volatile storage area 1168.

[0065] Data/information generated or captured by the mobile computing device 1100 and stored via the system 1102 may be stored locally on the mobile computing device 1100, as described above, or the data may be stored on any number of storage media that may be accessed by the device via the radio interface layer 1172 or via a wired connection between

the mobile computing device 1100 and a separate computing device associated with the mobile computing device 1100, for example, a server computer in a distributed computing network, such as the Internet. As should be appreciated such data/information may be accessed via the mobile computing device 1100 via the radio interface layer 1172 or via a distributed computing network. Similarly, such data/information may be readily transferred between computing devices for storage and use according to well-known data/information transfer and storage means, including electronic mail and collaborative data/information sharing systems.

[0066] Aspects of the present disclosure, for example, are described above with reference to block diagrams and/or operational illustrations of methods, systems, and computer program products according to aspects of the disclosure. The functions/acts noted in the blocks may occur out of the order as shown in any flowchart. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

[0067] The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the disclosure as claimed in any way. The claimed disclosure should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general inventive concept embodied in this application that do not depart from the broader scope of the claimed disclosure.

CLAIMS

What is claimed is:

1. A method for assisting users in re-finding documents, the method comprising:
generating embeddings for document clips related to respective documents among a plurality of documents;
receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;
generating embeddings for respective concept clips in the first set of concept clips;
generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips;
determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;
generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and
causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.
2. The method of claim 1, wherein the plurality of documents comprises a plurality of web pages in a browsing history of the user.
3. The method of claim 1, wherein the document clips related to content of a particular document among the plurality of documents include one or more of i) one or more text paragraphs parsed out from the particular document and ii) one or more images from the particular document.
4. The method of claim 1, wherein the document clips related to content of a particular document include at least one user-defined clip that includes information not extracted from the document.

5. The method of claim 1, wherein the first set of clips defining the first concept includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user.
6. The method of claim 1, wherein generating respective embeddings for the document clips comprises generating the embeddings for the document clips using a machine learning model configured to generate embeddings based on content, wherein the embeddings capture semantics of the content, and wherein generating embeddings for respective concept clips comprises generating the embeddings for the respective concept clips using the machine learning model.
7. The method of claim 1, wherein generating the first concept embedding comprises generating the first concept embedding based on a weighted combination of the embeddings generated for the respective concept clips in the first set of concept clips.
8. The method of claim 1, wherein generating the graphical user interface includes placing representations of the document clips in a multidimensional canvas, wherein the representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips.
9. The method of claim 8, wherein generating the graphical user interface further comprises:
 - placing a representation of the first concept at a first location in the multidimensional canvas;
 - determining locations of the representations of the document clips relative to the first location of the representation of the first concept based on calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding;
 - and
 - positioning the representations of the document clips in relation to the representation of the first concept in the multidimensional canvas in accordance to the determined distances.

10. The method of claim 9, further comprising:

receiving a second set of concept clips defining a second concept for searching for content of interest to the user in the plurality of documents;

placing a representation of the second concept at a second location in the multidimensional canvas; and

updating locations of the representations of the document clips such that i) representations of document clips that are more semantically similar to the first concept than the second concept are placed closer to the representation of the first concept in the multidimensional canvas and ii) representations of document clips that are more semantically similar to the second concept than the first concept are placed closer to representation of the second concept in the multidimensional canvas.

11. The method of claim 10, wherein updating locations of the representations of the document clips comprises:

generating embeddings for respective concept clips in the second set of concept clips;

generating a second concept embedding based on a combination of the embeddings generated for the respective concept clips in the second set of concept clips;

determining updated locations of the representations of the document clips in the multidimensional canvas based on i) the first location of the representation of the first concept in the multidimensional canvas, ii) calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding, iii) the second location of the representation of the second concept in the multidimensional canvas and iv) calculated distances between the embeddings generated for the corresponding document clips and the second concept embedding; and

moving the representation of the document clips to the updated locations in the multidimensional canvas.

12. The method of claim 10, further comprising:

detecting that the representation of the first concept has been repositioned to a new location in the multidimensional canvas by the user; and

in response to detecting that the representation of the first concept has been repositioned, repositioning representations of the document clips in the multidimensional canvas such that representations of document clips that are more semantically similar to the first concept than the second concept move closer to the new location of the representation of the first concept.

13. A system for assisting users in re-finding documents, the system comprising:
 - one or more computer readable storage media; and
 - program instructions stored on the one or more computer readable storage media that, when executed by at least one processor, cause the at least one processor to:
 - generate, using a machine learning model, embeddings for document clips related to respective documents among a plurality of documents;
 - receive a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;
 - generate, using the machine learning model, embeddings for respective concept clips in the first set of concept clips;
 - generate a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips;
 - determine semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;
 - generate a graphical user interface depicting the semantic relationships between the first concept and the document clips; and
 - cause display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

14. The system of claim 13, wherein the plurality of documents comprises a plurality of web pages in a browsing history of the user.

15. The system of claim 13, wherein the document clips related to content of a particular document among the plurality of documents include one or more of i) one or more text paragraphs parsed out from the particular document ii) one or more images from the particular document and iii) at least one user-defined clip that includes information not extracted from the document.

16. The system of claim 13, wherein the first set of clips defining the first concept includes one or more of i) one or more custom text paragraphs provided by the user, ii) one or more text paragraphs pasted from one or more documents by the user and iii) one or more images provided by the user.

17. The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the first concept embedding based on a weighted combination of the embeddings generated for the respective concept clips in the first set of concept clips.

18. The system of claim 13, wherein the instructions, which when executed by the processor, cause the processor to generate the graphical user interface at least by:

placing representations of the document clips in a multidimensional canvas in the graphical user interface, wherein the representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips;

placing a representation of the first concept at a first location in the multidimensional canvas;

determining locations of the representations of the document clips relative to the first location of the representation of the first concept based on calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding; and

positioning the representations of the document clips in relation to the representation of the first concept in the multidimensional canvas in accordance to the determined distances.

19. A computer storage medium comprising computer executable instructions, which when executed by at least one processor, perform a method comprising:

generating embeddings for document clips related to respective documents among a plurality of documents;

receiving a first set of concept clips defining a first concept for searching for content of interest to a user in the plurality of documents;

generating embeddings for respective concept clips in the first set of concept clips;

generating a first concept embedding based on a combination of the embeddings generated for the respective concept clips in the first set of concept clips;

determining semantic relationships between the first concept and the document clips based on i) the embeddings generated for the document clips and ii) the concept embedding;

generating a graphical user interface depicting the semantic relationships between the first concept and the document clips; and

causing display of the graphical user interface to be rendered at a client device, wherein the graphical user interface is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.

20. The computer storage medium of claim 19, wherein the method further comprises:

placing representations of the document clips in a multidimensional canvas in the graphical user interface, wherein the representations of the document clips are placed relative to each other based on calculated distances between the embeddings generated for the corresponding document clips;

placing a representation of the first concept at a first location in the multidimensional canvas;

determining locations of the representations of the document clips relative to the first location of the representation of the first concept based on calculated distances between the embeddings generated for the corresponding document clips and the first concept embedding; and

positioning the representations of the document clips in relation to the representation of the first concept in the multidimensional canvas in accordance to the determined distances.

ABSTRACT

A document re-finding system generates embeddings for concept clips provided by a user, the concept clips defining a concept for searching for content of interest to the user in a plurality of documents previously seen by the user. The re-finding system determines semantic relationships between the concept and document clips, related to respective document among the plurality of documents, based on the concept embedding and embeddings generated for the document clips. A graphical user interface depicting the semantic relationships is rendered to the user and is operable to enable re-finding a document, among the plurality of documents, having the content of interest to the user.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/541,216

RECEIPT DATE / TIME
12/15/2023 11:52:05 AM Z ET

ATTORNEY DOCKET #
MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 4414

FILED BY Irina Mikitiouk

PATENT CENTER # 63629944

FILING DATE -

CUSTOMER # 141674

FIRST NAMED INVENTOR Gonzalo A. RAMOS

CORRESPONDENCE ADDRESS -

AUTHORIZED BY Ethan Bell

Documents

TOTAL DOCUMENTS: 7

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
MST-0561_410864-US-CNT_ADS.pdf	9	Application Data Sheet	2174 KB
MST-0561_410864-US-CNT_Declarations.pdf	10	Oath or Declaration filed	1095 KB
MST-0561_410864-US-CNT_Patent_Application.pdf	39	-	253 KB
MST-0561_410864-US-CNT_Patent_Application-SPEC.pdf	(1-32) 32	Specification	210 KB
MST-0561_410864-US-	(33-38) 6	Claims	108 KB

CNT_Patent_Application-
CLM.pdf

MST-0561_410864-US- CNT_Patent_Application- ABST.pdf	(39-39)	1	Abstract	73 KB
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MST-0561_410864-US- CNT_Drawings.pdf		12	Drawings-only black and white line drawings	571 KB
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MST-0561_410864-US- CNT_Power_of_Attorney.pdf		2	Power of Attorney	237 KB
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Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

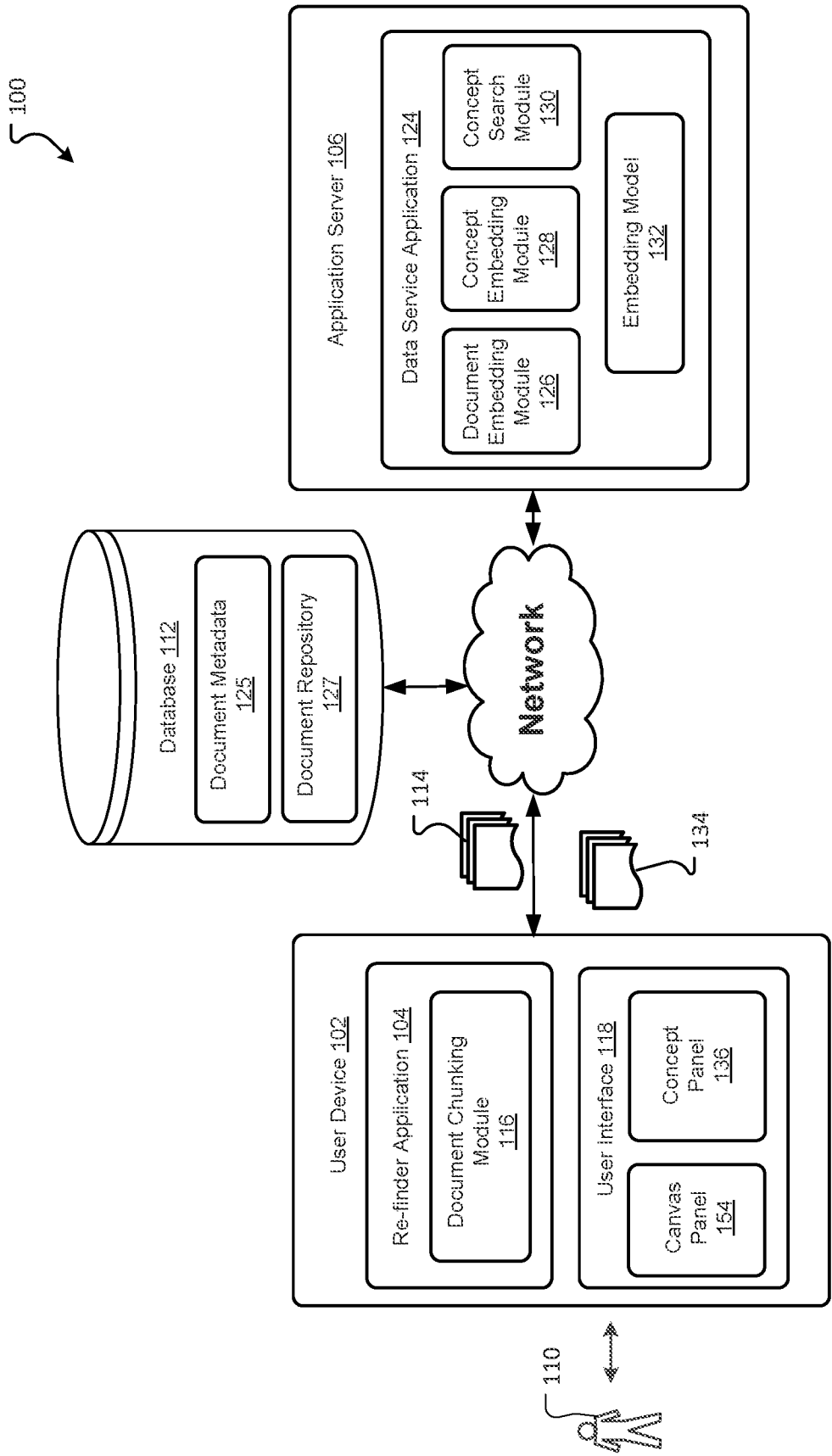


Fig. 1

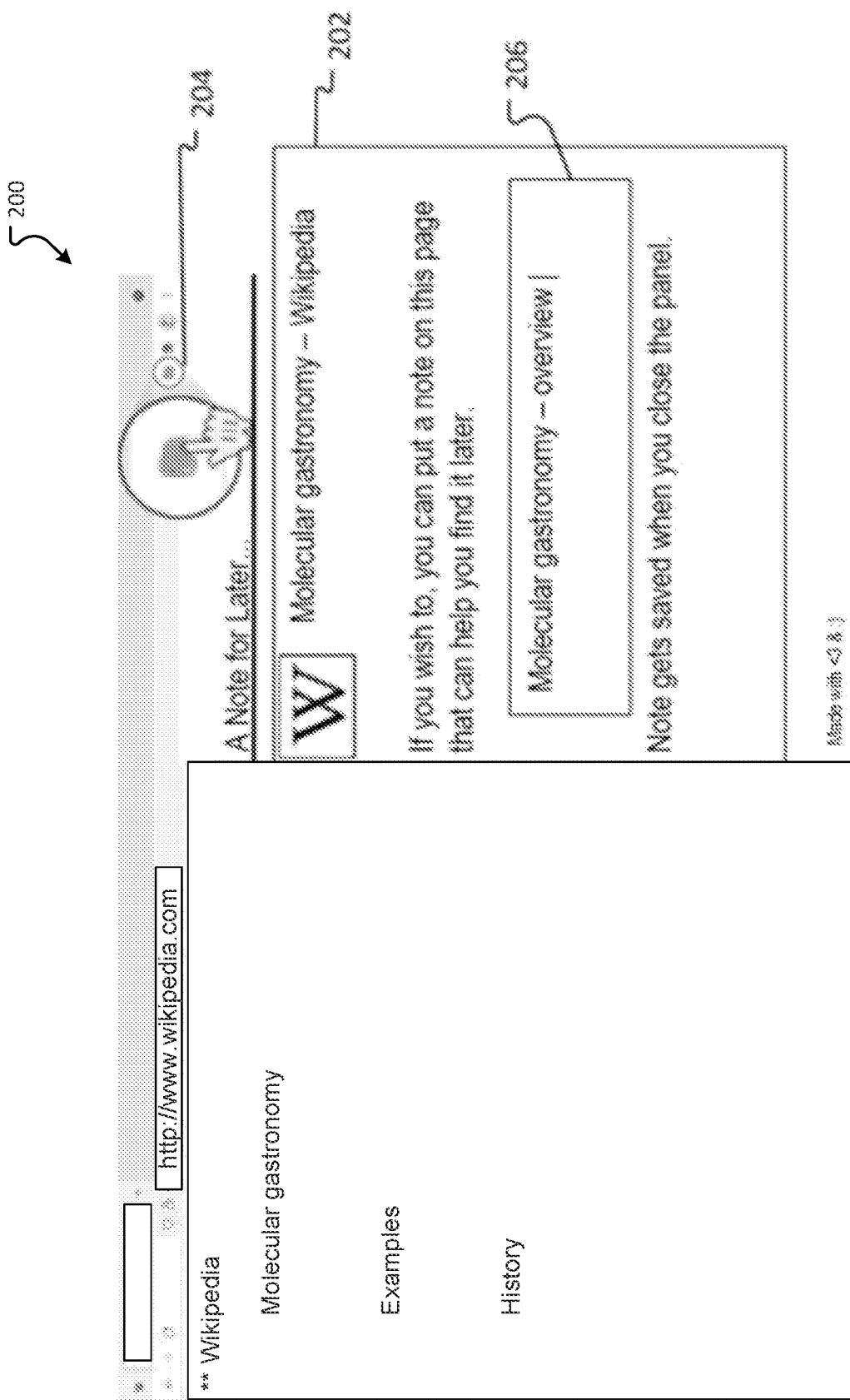


Fig. 2

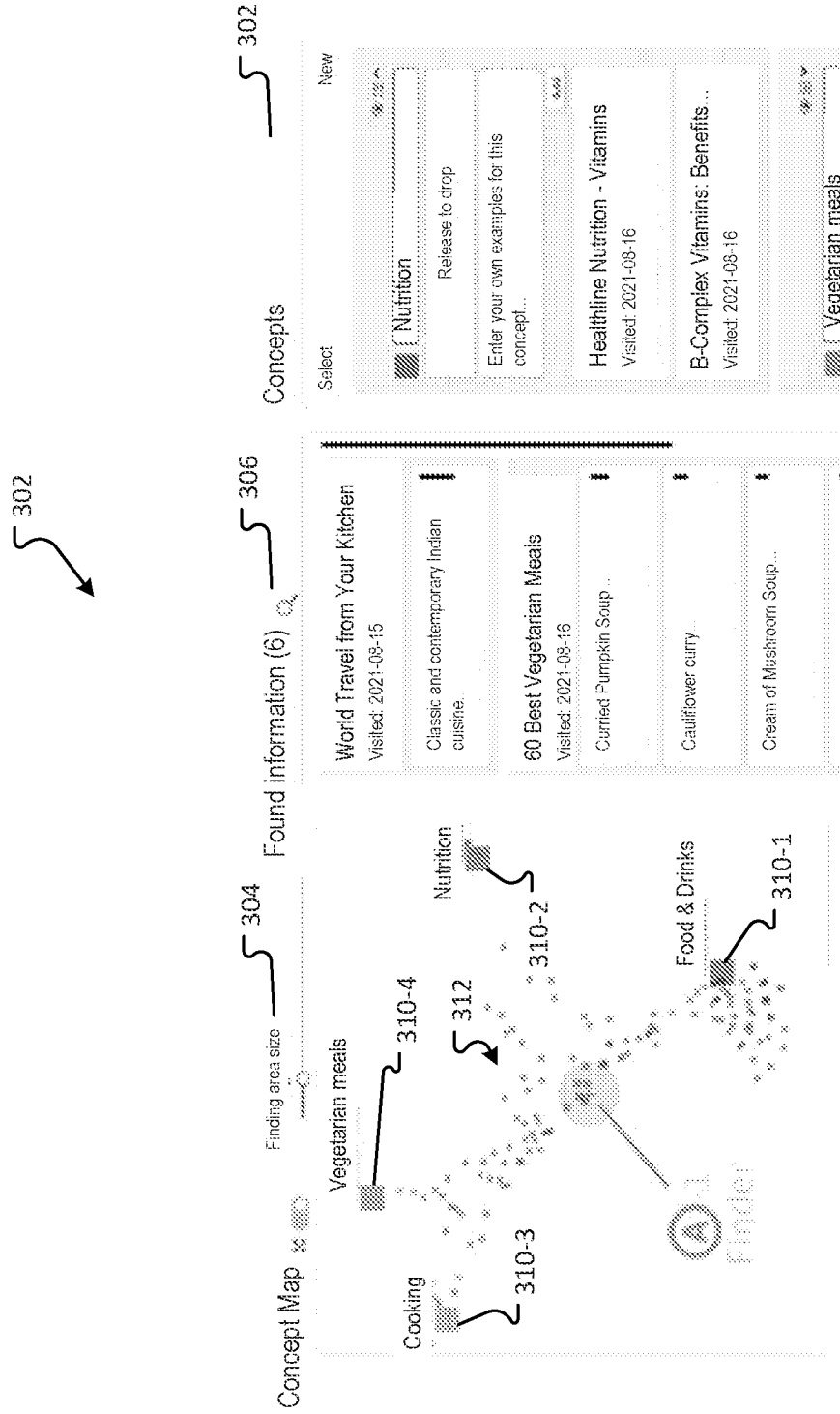


Fig. 3

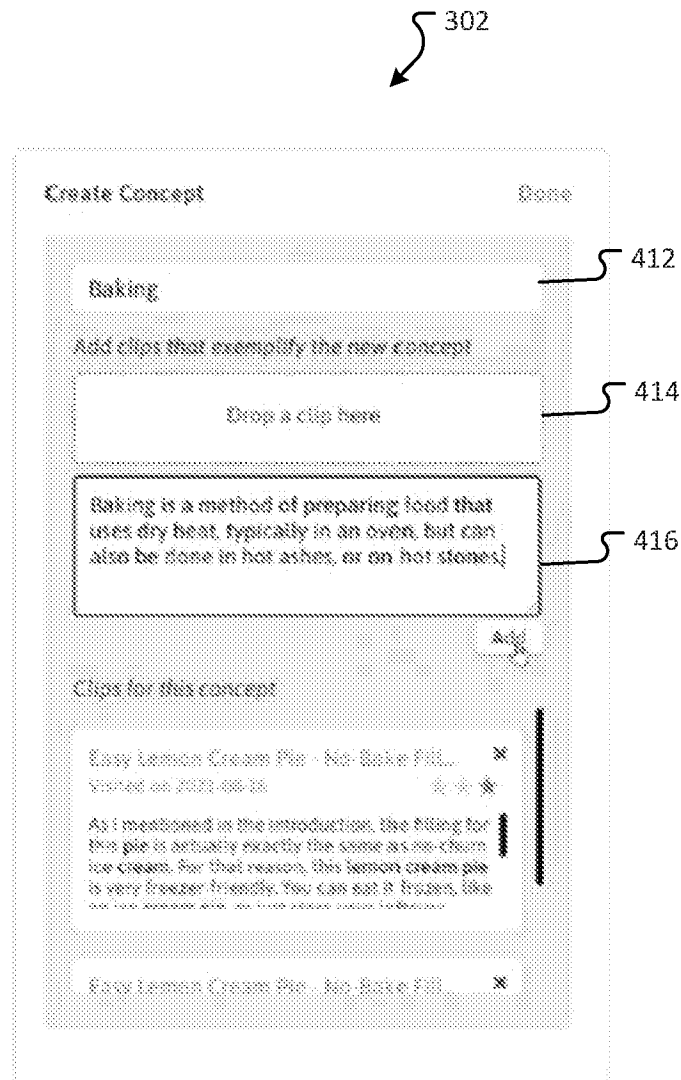


Fig. 4

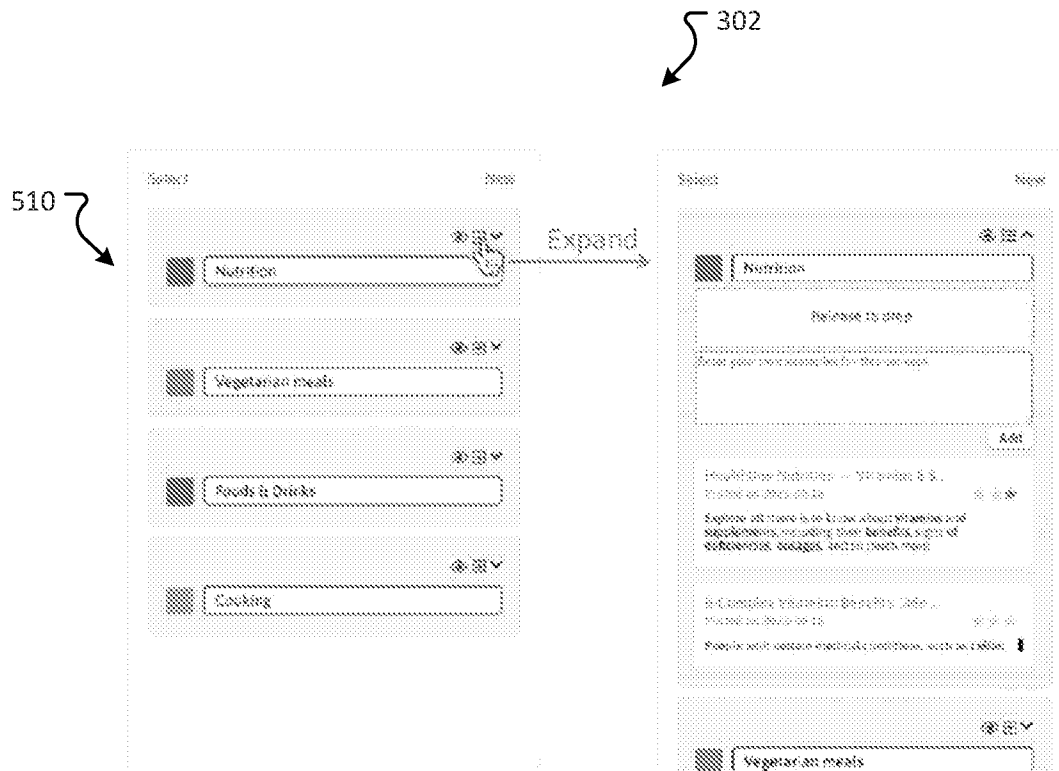


Fig. 5

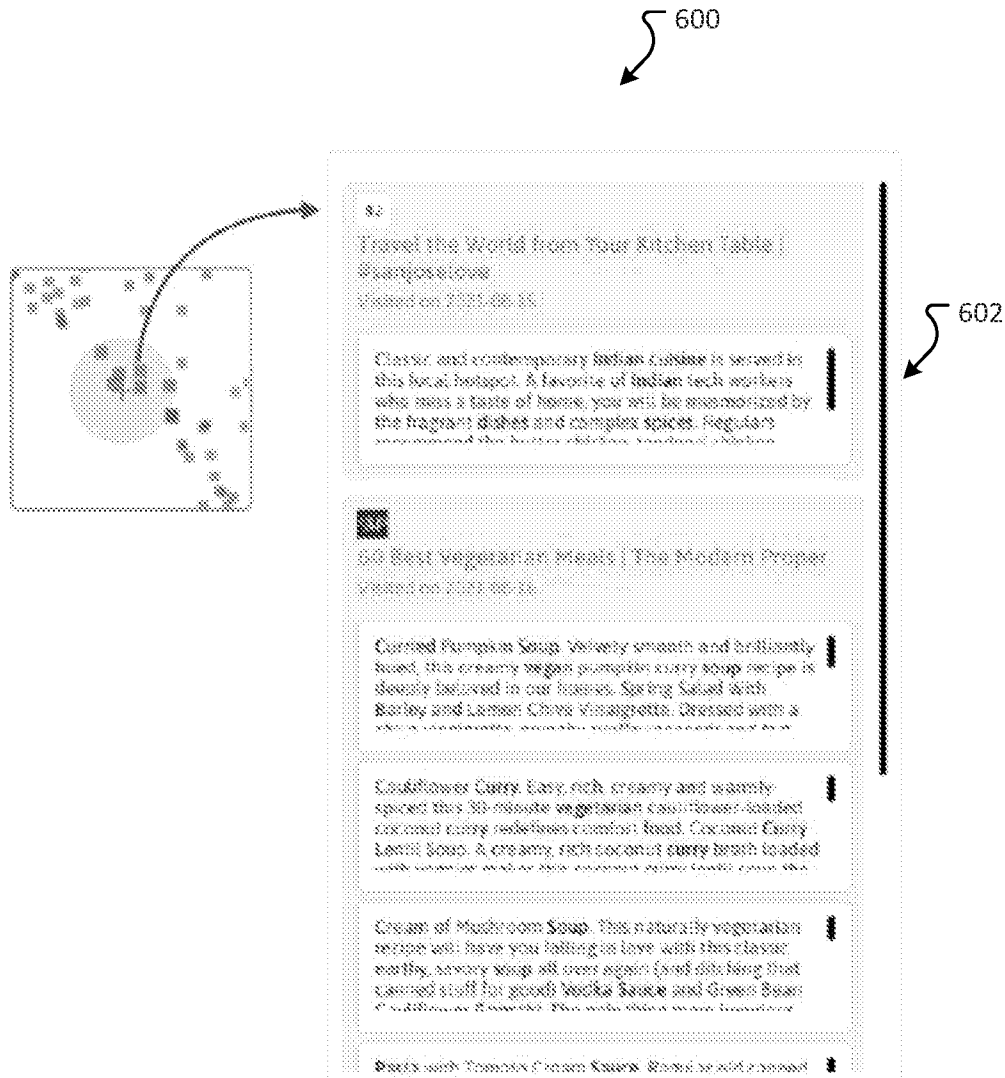
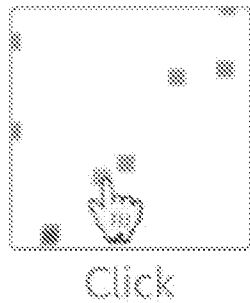


Fig. 6

622-1 ↗

↖ 700



↖ 702

Selected clip

Vegetable - Wikipedia
Visited on 2021-08-16

Before the advent of agriculture, humans were hunter-gatherers. They foraged for edible fruit, nuts, stems, leaves, corms, and tubers, scavenged for dead animals and hunted living ones for food [10] Forest gardening is a knowledge base, the structure is illustrated by the following:

Similar clips found by a language model

W
Vegetable - Wikipedia
Visited on 2021-08-16

Originally, vegetables were collected from the wild by hunter-gatherers and entered cultivation in several parts of the world, probably during the period 10,000 BC to 7,000 BC, when a new classification system of food developed for food security.

Books, reports, journals An Essay on Abstinence from Animal Food, as a Moral Duty (1802) Vegetable Cookery (1812) A Vindication of Natural Diet (1813) Reasons for not Eating Animal Food (1814) Moral Inquiries into the Duties of the Human Race (1814)

An Essay on Abstinence from Animal Food, as a Moral Duty (1802) Vegetable Cookery (1812) A Vindication of Natural Diet (1813) Reasons for not Eating Animal Food (1814) Moral Inquiries into the Duties of the Human Race (1814)

Fig. 7



Fig. 8

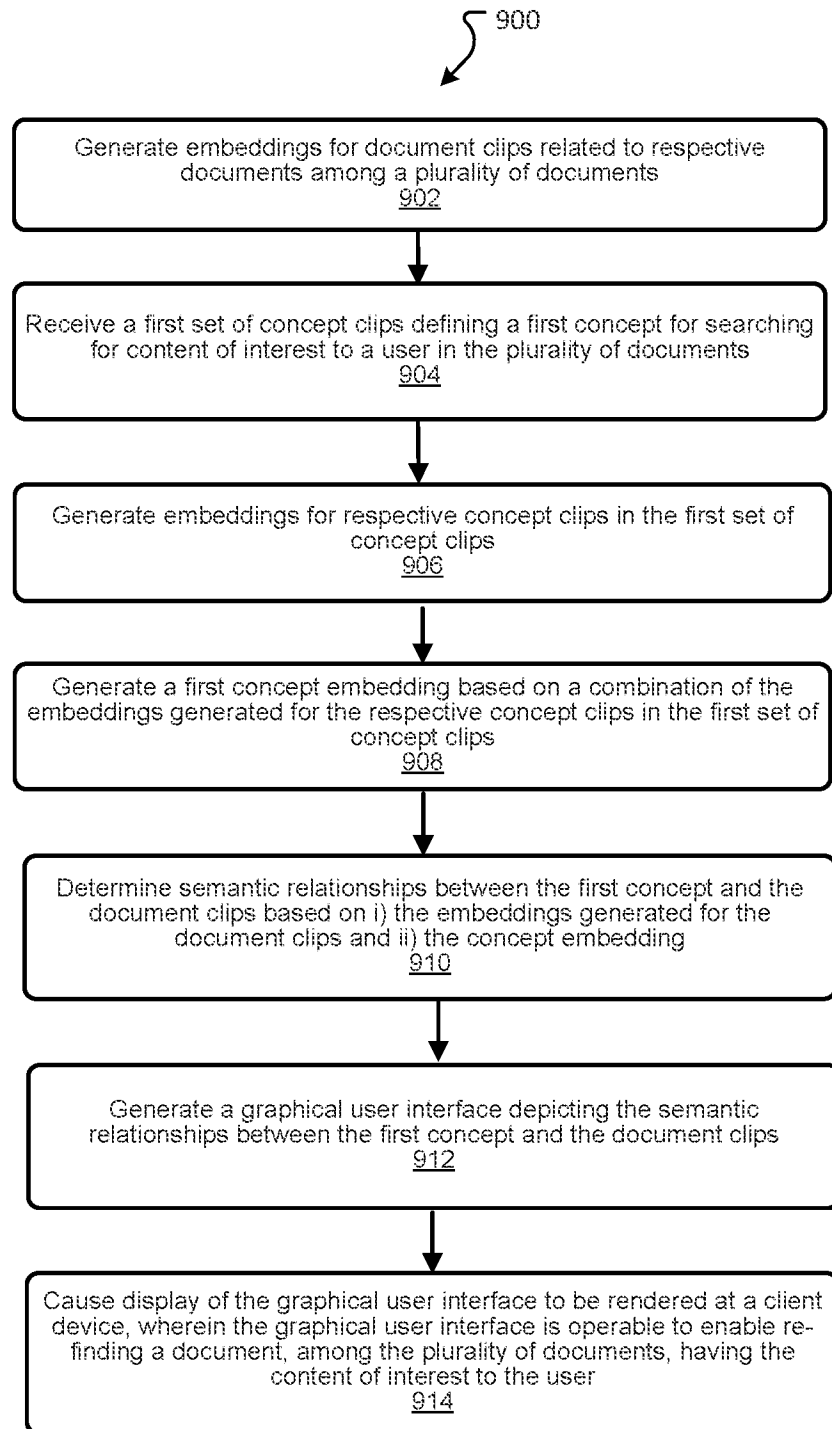


Fig. 9

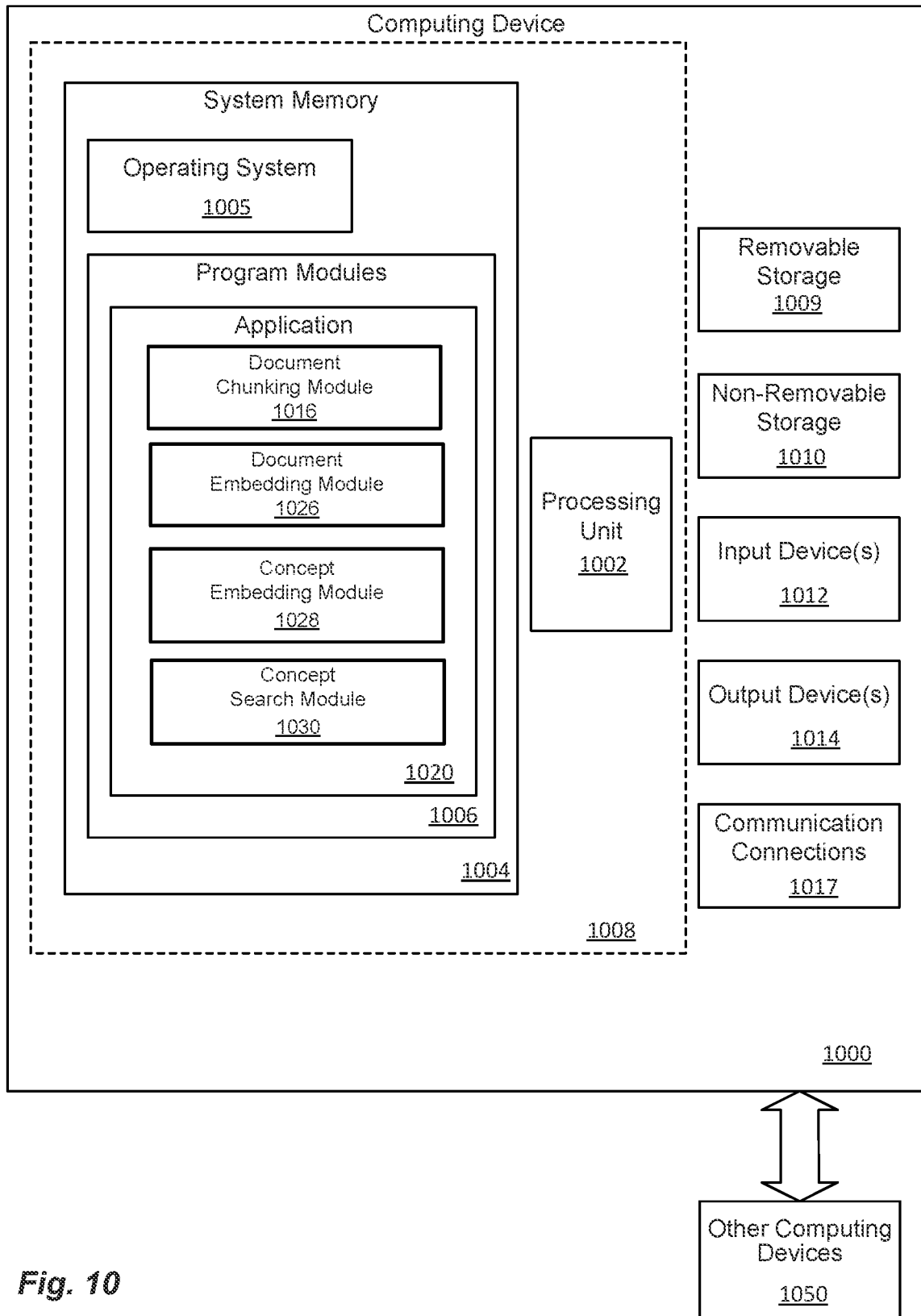


Fig. 10

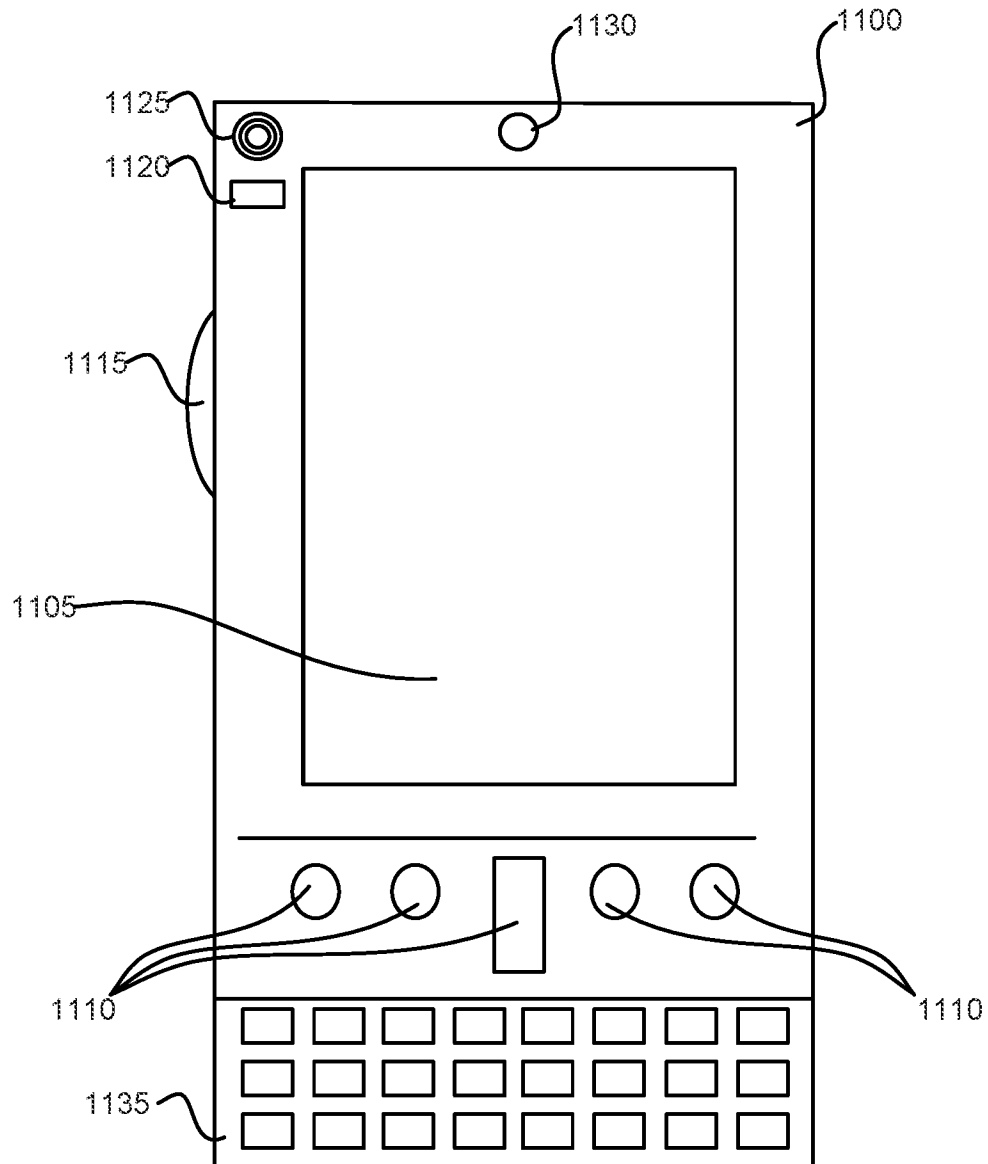


Fig. 11A

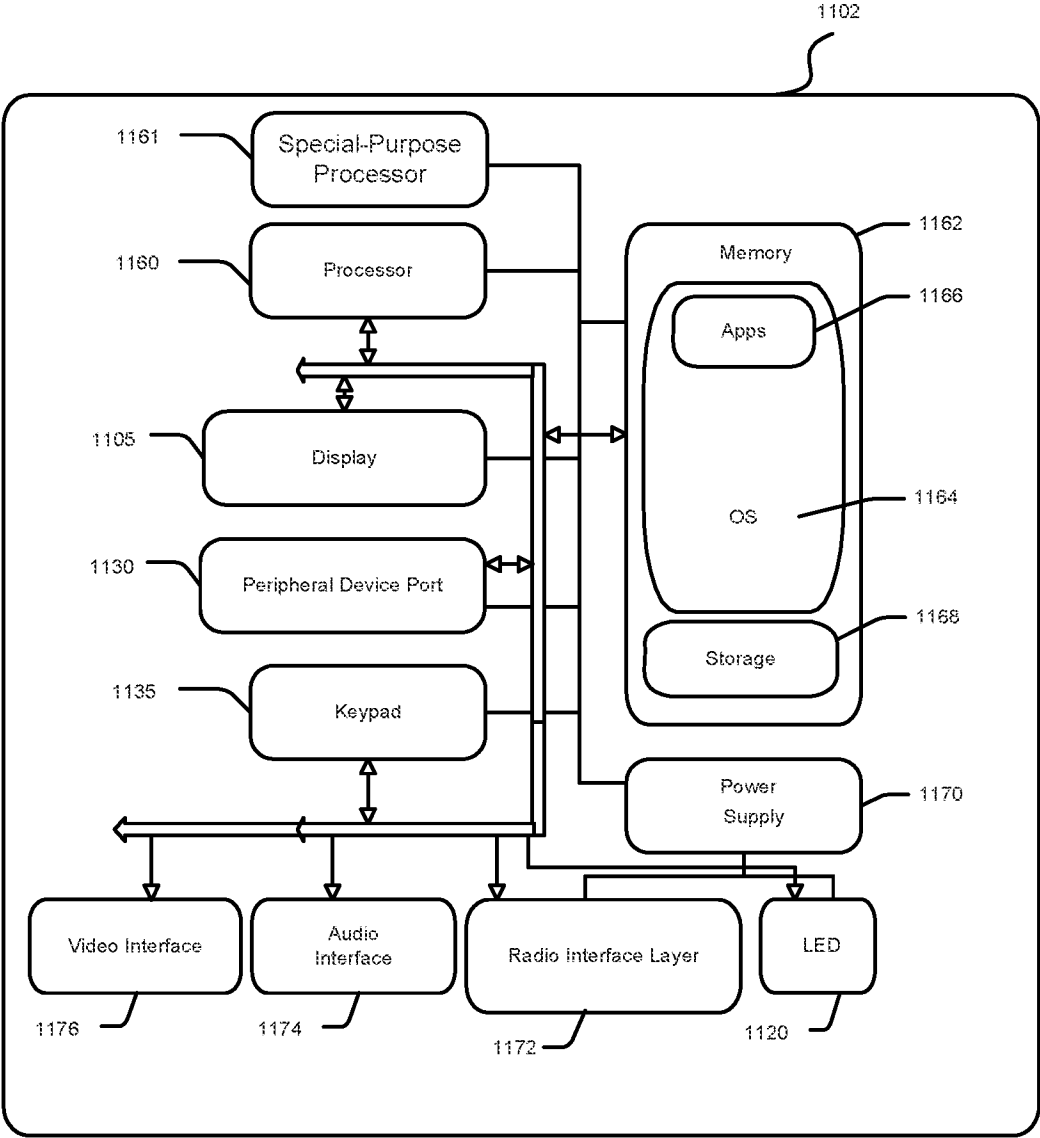


Fig. 11B



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ELECTRONIC PAYMENT RECEIPT

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RECEIPT DATE / TIME
12/15/2023 11:52:05 AM Z ET

ATTORNEY DOCKET #
MST-0561/410864-US-CNT

Title of Invention

DOCUMENT RE-FINDING SYSTEM

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 4414

FILED BY Irina Mikitiouk

PATENT CENTER # 63629944

AUTHORIZED BY Ethan Bell

CUSTOMER # 141674

FILING DATE -

CORRESPONDENCE
ADDRESS -

FIRST NAMED
INVENTOR Gonzalo A. RAMOS

Payment Information

PAYMENT METHOD
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PAYMENT TRANSACTION ID
E2023BEB53197523

PAYMENT AUTHORIZED BY
Irina Mikitiouk

PRE-AUTHORIZED ACCOUNT
500573

PRE-AUTHORIZED CATEGORY

37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.21 (Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1111	UTILITY PATENT APPL. SEARCH FEE	700.00	1	700.00
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T))	320.00	1	320.00
1311	PATENT APPL. EXAMINATION FEE	800.00	1	800.00
TOTAL AMOUNT:				\$1,820.00

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New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA/82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	Unassigned
Filing Date	Herewith
First Named Inventor	Gonzalo A. RAMOS
Title	DOCUMENT RE-FINDING SYSTEM
Art Unit	Unassigned
Examiner Name	Unassigned
Attorney Docket Number	MST-0561/410864-US-CNT

SIGNATURE of Applicant or Patent Practitioner

Signature	/Ethan W. Bell/	Date (Optional)	December 15, 2023
Name	Ethan W. Bell	Registration Number	75,549
Title (if Applicant is a juristic entity)	Attorney for Applicant		
Applicant Name (if Applicant is a juristic entity)	MICROSOFT TECHNOLOGY LICENSING, LLC		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



*Total of 1 forms are submitted.

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POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

141674

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number: 141674

OR



Firm or
Individual Name

Address

City

State

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Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Microsoft Technology Licensing, LLC



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being used with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

Name

Benjamin C. Orrison

Title

Secretary, Microsoft Technology Licensing, LLC

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of

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